

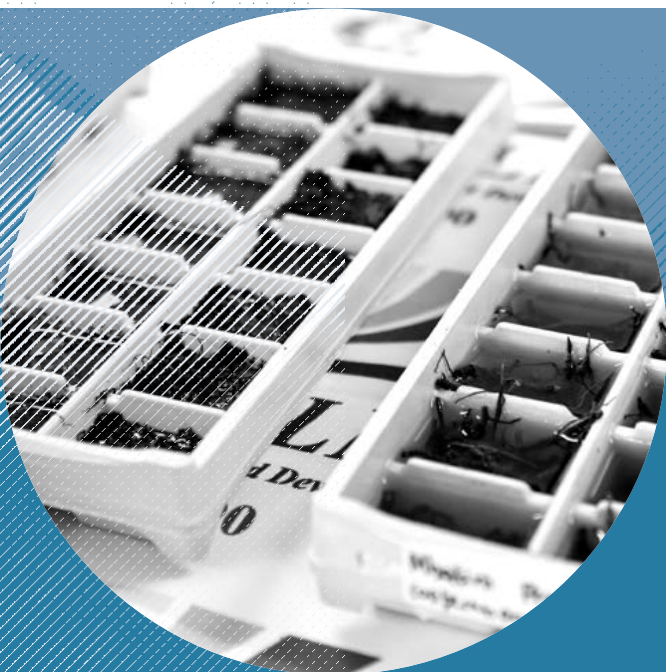


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SOIL HEALTH AND FERTILIZER POLICY AND INVESTMENT PROSPECTS IN SUB-SAHARAN AFRICA



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SOIL HEALTH AND FERTILIZER POLICY AND INVESTMENT PROSPECTS IN SUB-SAHARAN AFRICA

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Published by:
The Food and Agriculture Organization of the United Nations
and
Agrinatura
and
The European Union
Rome, Paris and Brussels, 2026

Required citation:

Jafari, A., Le Cotty, T. & Tefft, J. 2026. *Soil health and fertilizer – Policy and investment prospects in sub-Saharan Africa*. Directions in Investment no. 19. Rome, Paris and Brussels, FAO, Agrinatura and the European Union. <https://doi.org/10.4060/cd9916en>

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ISBN 978-92-5-140746-2

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Art direction and design: Sonia Malpeso & Eduardo Arenas

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Preface

This technical report was produced in 2025, prior to the closure of the Strait of Hormuz in 2026.^{1, 2}

The report grew out of a shared recognition among the European Union, FAO, and Agrinatura that a fresh approach is needed – since decades of subsidy programmes, research and investment have achieved only limited success in improving soil health and fertilizer use in sub-Saharan Africa.

It sets out policy options and decisions that need to be made, and is aimed at policymakers, development practitioners, and researchers working on sustainable food systems transformation across Africa.

The hope is that it will also prove useful to development finance institutions and others focused on agrifood systems in the region so they can deliver sustainable improvements in production, food security and nutritional outcomes.

The report aligns with, and was partly shaped by, the 2024 African Union Nairobi Declaration: Boosting Africa's Agricultural Productivity with Healthy Soils – a political commitment that reflects the need to address these issues at the highest levels.

Since the report's completion, the closure of the Strait of Hormuz has introduced new and significant risks to global energy, fertilizers, and food systems, with particular implications for Africa as a region that relies on imports for much of its energy and fertilizer needs.

Supply disruptions resulting from the conflict, affecting oil, gas, and fertilizer trade, have already triggered substantial price spikes, with significant implications for the sub-Saharan smallholder farmers at the heart of this report.

¹ For further information on the impacts of recent developments, please refer to the FAO brief *Global agrifood implications of the 2026 conflict in the Middle East*, produced by the FAO Chief Economist's Office, which analyses impacts on energy and fertilizer trade and food security. Additional FAO analytical publications address other dimensions of these developments.

² FAO. 2026a. *Global agrifood implications of the 2026 conflict in the Middle East: impacts on energy and fertilizer trade and food security*. Rome, FAO. [Cited 4 May 2026]. <https://openknowledge.fao.org/handle/20.500.14283/cd8875en>

Acknowledgements

This report was researched and written by Areej Jafari (FAO), Tristan Le Cotty (CIRAD), and James Tefft (FAO).

It was produced in the context of the Sustainable Agrifood Systems Intelligence (SASI) initiative, a collaboration between FAO, the European Union and Agrinatura, aiming to improve the availability of knowledge and guidance, globally and nationally, on the institutions, policies and investments needed to accelerate the transition to sustainable agrifood systems.

Its preparation greatly benefited from the valuable comments, reviews, and technical contributions provided by colleagues from the European Commission's Directorate General for International Partnerships (DG INTPA), Sustainable Agri-Food Systems and Fisheries Unit, European universities, and numerous experts across FAO technical divisions and decentralized offices.

The authors would like to express their sincere appreciation to all those who contributed to this report. Special thanks to colleagues from EU DG INTPA: Philippe Thomas, Christophe Larose, Guy Faure, Nicoletta Avella, Philippe Le Clerc; CIRAD: Thierry Brunelle, Gatien Falconnier, François Affholder, Eric Scopel, Elodie Maître d'Hôtel, Jean-Michel Sourisseau, Sylvain Perret, Marc Corbeels, Jean-Paul Laclau, Eric Justes; Swedish University of Agricultural Sciences (SLU): Konstantinos Karantininis; University of Porto (ISPUP): Gabriela Albuquerque; and Natural Resources Institute (NRI): Chris Turner.

We also thank our FAO colleagues for their valuable contributions:

- FAO Chief Economist's Office, led by Máximo Torero, and his team
- Plant Production and Protection Division: Eid Mohammed, Ronnie Brathwaite, Chikelu Mba, Yurdi Yasmi
- Land and Water Division: Lifeng Li, Nora Berrahmouni, Yuxin Tong, Vinisa Saynes Santillan, Imane Slimani
- FAO Regional Office for Africa: Valere Nzeyimana, Abebe Haile-Gabriel
- FAO Sub Regional Office of West Africa: Eugene Rurangwa, Adin Bloukounon Goubalan
- Trade and Market Division: Delphine Leconte-Demarsy, Monika Tothova, Maria Antip
- FAO Regional Office for Asia and Pacific: Meeta Punjabi Mehta
- FAO Investment Centre: Junko Nakai, Jaap van de Pol, Dario Lucantoni, Valeria Casavola, Kanika Garg, Alexandra Gonzalez, Nuno Santos, Béatrice Ngirabacu, and Mohamed Manssouri.

The authors also wish to thank the editorial and publication team of the FAO Investment Centre and especially Clare O'Farrell and Luigi Baldassari for coordinating the review and production of the report, Rex Merrifield for the editorial work, and Sonia Malpeso for the design creation and, together with Eduardo Arenas, the design work.

The authors are deeply grateful for the time, expertise, and thoughtful insights provided by all contributors, which significantly enhanced the quality and relevance of this report.

Abbreviations

AATF	African Agricultural Technology Foundation
AfDB	African Development Bank
AFFM	Africa Fertilizer Financing Mechanism
AFSH	Africa Fertilizer and Soil Health (Summit / Action Plan)
AGF	African Guarantee Fund
AGRA	Alliance for a Green Revolution in Africa
ASTI	Agricultural Science and Technology Indicators
AU	African Union
AUC	African Union Commission
AUDA-NEPAD	African Union Development Agency – New Partnership for Africa's Development
CAADP	Comprehensive Africa Agriculture Development Programme
CASA	Commercial Agriculture for Smallholders and Agribusiness
CGIAR	Consultative Group on International Agricultural Research (now One CGIAR)
CIRAD	French Agricultural Research Centre for International Development
DFI	development finance institution
ECOWAS	Economic Community of West African States
EU	European Union
FAGACE	African Guarantee and Economic Cooperation Fund
FAO	Food and Agriculture Organization of the United Nations
FCDO	Foreign, Commonwealth and Development Office (United Kingdom)
FOB	free on board
FSA	African Solidarity Fund
GDP	gross domestic product
GSP	Global Soil Partnership
HLPE	High Level Panel of Experts on Food Security and Nutrition
IEA	International Energy Agency
IFA	International Fertilizer Association
IFAD	International Fund for Agricultural Development
IFDC	International Fertilizer Development Center
IFPRI	International Food Policy Research Institute
IITA	International Institute of Tropical Agriculture
ISFM	integrated soil fertility management
ISRIC	International Soil Reference and Information Centre
ITPS	Intergovernmental Technical Panel on Soils
LEISA	low-external-input sustainable agriculture
MIRCEN	Microbiological Resources Centre
NEPAD	New Partnership for Africa's Development

NGO	non-governmental organization
NPK	nitrogen–phosphorus–potassium (compound fertilizer)
R&D	research and development
SASI	Sustainable Agrifood Systems Intelligence
SOC	soil organic carbon
SOFI	The State of Food Security and Nutrition in the World
SoilFER	Soil Mapping for Resilient Agrifood Systems Programme
SSA	sub-Saharan Africa
TFP	total factor productivity
UN	United Nations
UNECA	United Nations Economic Commission for Africa
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WFO	World Farmers' Organisation
WFP	World Food Programme
WHO	World Health Organization
WTO	World Trade Organization

Executive summary

Agriculture is a cornerstone of the economy of sub-Saharan Africa (SSA), contributing 17.5 percent of the region's gross domestic product (GDP) and employing more than 60 percent of the population. **Yet the sector faces complex and persistent challenges** (World Bank, 2025a), and falls short of meeting the food security needs of the continent. A fifth of the SSA population faces hunger – the highest percentage globally (FAO et al., 2025) – and land degradation affects 65 percent of the productive land. Productivity remains low, particularly on small farms where most food is produced. Although total output has increased over time, this growth has been driven primarily by converting forest and natural land into farmland, rather than by productivity gains. These dynamics have reduced soil carbon storage capacity and diminished soil health, compounding long-term threats to food security. They undermine agricultural productivity and trap smallholder farmers in a vicious cycle of poor yields, limited resources, and deteriorating soil conditions.

Fertilizers are a key factor in breaking this cycle, yet their efficient use in SSA remains low due to structural, market and agronomic constraints. Average nitrogen fertilizer use stands at 13.3 kg/ha – roughly a fifth of the global average of 65.4 kg/ha (FAO, 2024a). High and variable fertilizer prices, driven largely by import dependence, are compounded by distribution challenges, inadequate infrastructure, high transaction and transportation costs, import administration costs and limited market competition. Supply-side constraints alone, however, do not fully explain low fertilizer use.

On the demand side, limitations include inadequate information on soil type, soil quality, farmers' perceptions of fertilizer quality, inadequate size packaging, ineffective agricultural extension, and limited access to financing, as well as marketing difficulties that contribute to unprofitability. Fertilizer efficiency is also undermined by poor soil fertility and health, lack of site-specific recommendations, and limited adoption of complementary approaches for sustainable intensification. These challenges, exacerbated by persistently high prevalence of poverty (estimated to range between 40 percent and 80 percent across different SSA countries), constrain effective demand for fertilizer. Underinvestment in agricultural research and development, along with weak regulatory frameworks – especially for promising options such as bio-based fertilizers³ – continue to narrow the available options.

Crucially, for subsistence farmers, low fertilizer use is often a rational response to risk rather than simply a matter of limited knowledge or weak incentives. Investments in soil health may compete directly with immediate food security needs of the family. Uncertain returns, climate variability, and weak output markets further increase the cost of failure. Constraints ranging from

³ Bio-based fertilizers refer to fertilizers derived from biological materials such as plant residues, animal manure, or microbial biomass (including biofertilizers and microbial byproducts). These products provide essential nutrients to plants and encompass a wide range of fertilizers of biological origin, including organic fertilizers such as compost and livestock manures. This term includes biofertilizers, which consist of living microorganisms such as bacteria or fungi that enhance plant nutrition by fixing atmospheric nitrogen or by mobilizing and increasing the availability of nutrients in the soil and plants (FAO, 2026b; Bharti and Suryavanshi, 2021). For more details on fertilizer types see Annex 2 and IFA (2024).



volatile prices, insecure land access and tenure, to climate risk and farm fragmentation interact and reinforce one another, meaning that progress in one area is frequently held back by limitations in others. Other rural constraints may include limited market access for surplus production, offering little scope to take advantage of growing urban food demand.

Empirical evidence shows that addressing fertilizer independently of other aspects contributing to soil health yields limited and short-lived results.

Where fertilizer has contributed to sustained agronomic productivity gains – such as in parts of Kenya, Ethiopia and Nigeria – it has rarely done so in isolation. Instead, gains have occurred in better-endowed agroecological zones or settings where farmers had reliable market access and complementary investments. Integrated, coordinated approaches are thus essential.

This report examines options for enhancing soil health and efficient fertilizer management among small-scale farmers in SSA – in alignment with the principles of the International Code of Conduct for the Sustainable Use and Management of Fertilizers (FAO, 2019), which serves as FAO's main framework for promoting responsible and sustainable fertilizer management globally. The analysis offers evidence-based insights that go beyond the existing polarized debates around fertilizers, to explore a range of policy options and actions aimed

This report examines options for enhancing soil health and efficient fertilizer management among small-scale farmers in SSA.

at mitigating further soil degradation, rehabilitating degraded land, and optimizing fertilizer use on both healthy and degraded lands. These perspectives aim to balance long-term soil health and food security needs of smallholder farmers with their short-term profitability, and the varying needs of soils and crops.

The main approaches to fertilizer application and soil health have evolved from the green revolution focus on high-yielding varieties and chemical inputs toward more sustainable, systemic approaches linking smallholder farmers, biological processes, and mineral fertilizers. This shift recognizes that while fertilizers are necessary to increase productivity growth, their effectiveness depends on proper application and soil conditions. In alignment with the 2024 Nairobi Declaration and the African Union's Fertilizer and Soil Health Action Plan, integrated soil fertility management (ISFM) has emerged as a way to optimize both bio-based and mineral nutrient use, with fertilizers as an entry point to sustainable intensification. Agroecology and regenerative agriculture have also gained prominence, highlighting soil health in enhancing agrifood systems sustainability, and leading to multiple co-benefits including climate resilience, biodiversity restoration, ecosystem services and food security.

For many resource-constrained smallholders, practices such as intercropping, crop rotation, and crop–livestock integration are not transitional steps toward input-intensive systems. They are core producer strategies for managing soil fertility and production risk – for which their feasibility is constrained by farm size, with very small plots often pushing farmers toward continuous monocropping that accelerates soil degradation.

In the context of the Africa Fertilizer and Soil Health Action Plan: 2024–2034, this report presents four priority policy options to improve soil health and sustainably boost food production as critical elements for the sustainable transformation of agrifood systems in sub-Saharan Africa. The 2025 Kampala CAADP Declaration, which sets out the Comprehensive Africa Agriculture Development Programme (CAADP) agenda for 2026–2035, reaffirms Africa's commitment to invest in restoring and maintaining soil fertility and health as a cornerstone of sustainable agrifood systems. Viewed together as part of an integrated soil health and fertilizer policy framework, these priority policy options consist of: (i) improving soil health and soil fertility; (ii) increasing the availability of fertilizers; (iii) increasing access to and affordability of quality fertilizers; and (iv) increasing producers' production incentives. The options are assessed against five criteria: technical efficiency, economic, environmental and social impact, as well as scalability.

Within the four policy priorities, the report assesses a diverse set of tools and options for action that stakeholders could consider and tailor to local contexts. Investment in rural infrastructure (roads, storage, information systems) and fostering market competition can increase availability and lower farm-gate fertilizer prices. Expanding local fertilizer production, including mineral fertilizer manufacturing and blending, green hydrogen-based fertilizer, bio-based fertilizer, can also enhance fertilizer availability. Policymakers can accompany these efforts with actions to incentivize farmers' demand for fertilizers and sustainable production practices, by addressing a range of constraints that smallholder farmers may face. Because constraints interact and differ systematically across farm types, instruments must be differentiated accordingly. For subsistence farmers, risk-reduction measures – crop insurance, credit access, and income buffers – may need to precede or accompany fertilizer promotion. For semi-commercial and commercial farmers, improved market access and site-specific recommendations are likely more binding constraints.

The evidence presented in this report thus points clearly away from one-size-fits-all approaches to assessing and prioritizing diverse tools and options. This may require governments to radically rethink fertilizer support programmes, moving away from strategies that treat smallholders as a uniform group. To avoid continued underperformance of such blanket methods, countries could consider more flexible, nuanced and comprehensive approaches along the lines of ISFM to meet the needs of farmers and their soils. Critically, the efficiency of fertilizer application depends highly on the agronomic conditions of soil health. Appropriate, relevant approaches could include integrating mineral fertilizers with ISFM and agroecological practices, such as adequate nutrient cycling, legume and water management infrastructure, crop-livestock integration, and biofertilizers. An important finding in the report is that investing in permanent soil health and infrastructure (e.g. production, market and roads) yields more sustainable long-term benefits than simply subsidizing fertilizers. Thus, in many settings, market integration and risk reduction will need to accompany or even precede fertilizer promotion if investments in soil health are to generate durable returns.

Transforming Africa's soil health and fertilizer management will require a multi-pronged approach to inform policy and catalyse investments. This will need sustained political commitment and coordinated investment through partnerships between farmers, governments, financial institutions, research centres, agribusinesses, and stakeholders from sectors beyond agriculture. A strong enabling environment – backed by coherent regulation, robust monitoring systems, and public-private partnerships – would also contribute to more sustainable agrifood systems in Africa.

The diverse policy tools outlined in this report address trade-offs that will need to be made. Short-term approaches alone cannot adequately feed growing populations, while long-term policies often encounter political constraints. Ideally, countries would define their long-term objectives for sustainable agrifood systems and design and implement policies to move towards these goals. The African Union and regional economic groupings can play a central role in holding member states accountable to the long-term agrifood system strategies and budgets this transformation demands. The analytical tools and policy frameworks to do this already exist – what remains is the institutional will to apply them at scale. The knowledge basis and the policy toolbox are ready for this journey.



Introduction

1.1 AGRIFOOD SYSTEMS AND SOIL HEALTH IN SUB-SAHARAN AFRICA: CONTEXT AND BACKGROUND

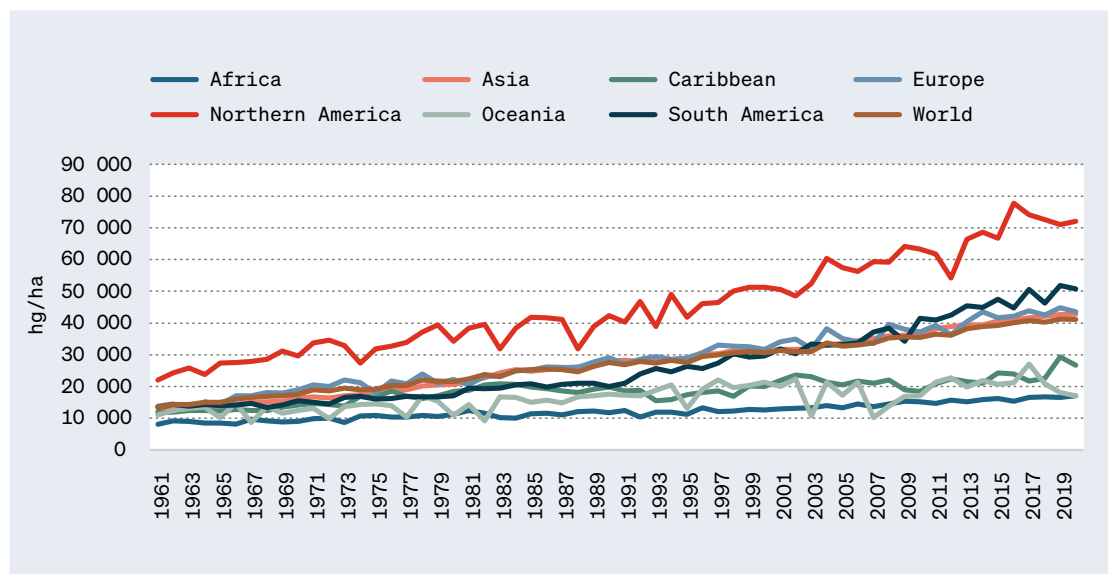
Sub-Saharan Africa continues to face the challenge of food insecurity. An estimated 307 million people living in the region experienced food insecurity in 2024, some 20.2 percent of the population – the highest proportion in the world (FAO *et al.*, 2025). The persistent lack of food security results from a complex interplay of factors, including high levels of poverty, low agricultural productivity, uneven structural transformation, poor infrastructure, weak governance, chronic conflict, soil degradation, climate change and weather shocks (CFS, 2021a; Vanlauwe *et al.*, 2023; Giller, 2020).

Agriculture continues to be the foundation of Africa's economy and contributed 17.5 percent of SSA's GDP in 2024 (World Bank, 2025a), with significantly higher contributions in countries such as Somalia (55.8 percent) and Niger (47.8 percent) (World Bank, 2025a). More than 60 percent of the region's population relies on agriculture for a livelihood, a figure rising to 80 percent in Somalia and 86 percent in Burundi (World Bank, 2022a). Approximately 70 percent of food production in low- and middle-income countries comes from farms smaller than 20 hectares (Woodhill, Hasnain and Griffith, 2020). Agricultural productivity-led growth is a driving force in rural and structural transformation in SSA, though varying in this role within and between countries (Gollin, 2023).

Traditional pathways for agricultural transformation face significant challenges in SSA, including constraints on agricultural productivity growth, farm consolidation and movement of labour to other sectors of the economy. These pathways are hindered by rural population growth, land fragmentation, and vulnerability to climate change, as well as a lack of value addition and off-farm employment opportunities – critical to raise incomes and enhance food security (Giller and Andersson, 2024).

FIGURE 1

Cereal yields over time, Africa and other regions, 1961–2020



SOURCE: FAO. 2024. FAOSTAT: Crops and livestock products. [Accessed 29 April 2026]. <https://www.fao.org/faostat/en/#data/QCL>. Licence: CC BY-4.0.

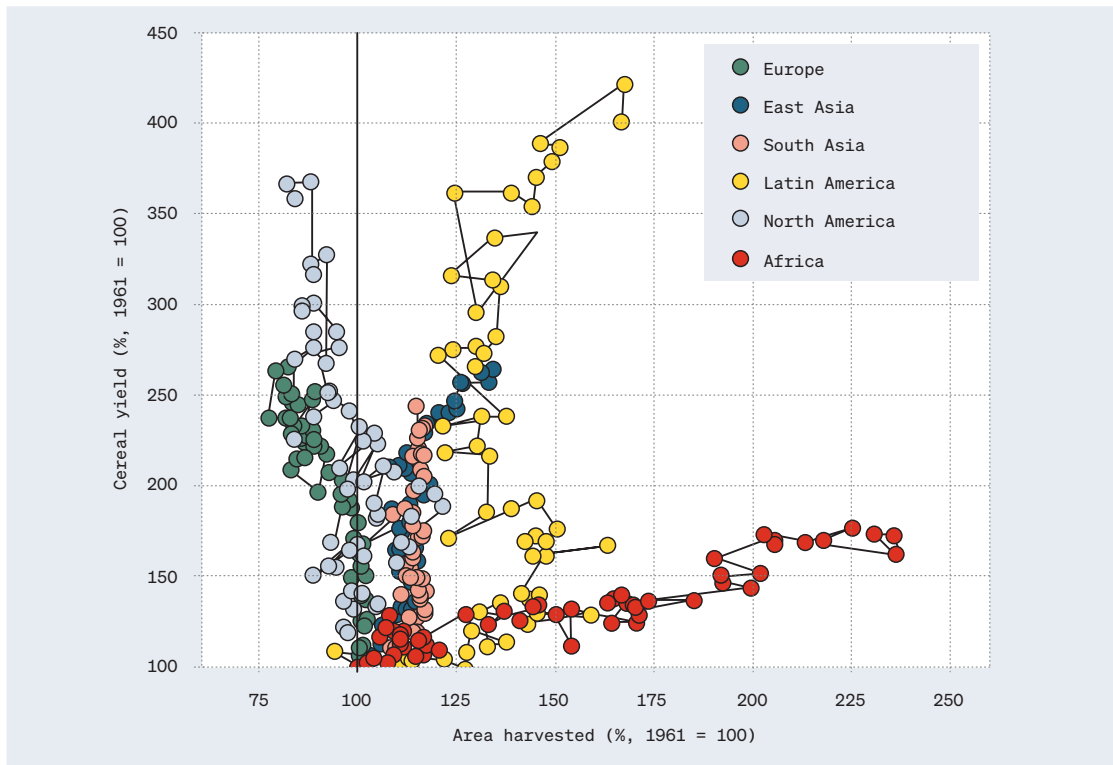
Agricultural productivity growth in Africa as a whole – and particularly in SSA – remains alarmingly low

(Fuglie et al., 2019). The continent lags the rest of the world, with cereal yields doubling from 0.81 tonnes/ha in 1961 to 1.7 tonnes/ha in 2020 – less than half of global cereal yields estimated at 4.18 tonnes/ha in 2020 (see Figure 1). Panel data (2008–2019) from smallholder farmers across six SSA countries show no significant improvement in crop total factor productivity (TFP) (Wollburg et al., 2024),⁴ despite increasing investments in African agriculture over the last 20 years. While total agricultural output has increased over time, this growth is driven primarily by land expansion (see Figure 2) rather than significant productivity gains (Giller et al., 2021). Further, extensification through conversion of forest and other natural land into farmland reduces soil capacity to store carbon, exacerbating land degradation and contributing to greenhouse gas emissions (Olsson et al., 2019; Ansari et al., 2022).

Africa's low agricultural productivity growth has been attributed to several challenges, such as limited input use, restricted access to quality seed, and heavy reliance on rainfed agriculture with insufficient irrigation infrastructure. These constraints are compounded by soil health issues, including nutrient mining, climate shocks, food price volatility, and insecure and customary land tenure (Jayne and Sanchez, 2021).

Small African economies struggle with limited capacity for technology adoption and adaptation, while underinvestment in agricultural R&D hinders the development of farming techniques suited to the continent's diverse agro-

⁴ TFP measures the efficiency with which inputs such as land, water, and fertilizers are combined to produce agricultural output, often through the adoption of improved technologies and practices. See: (Fuglie et al., 2019)

FIGURE 2**Intensification and area expansion trajectories for different regions, 1961-2020**

SOURCE: Giller, K.E., Delaune, T., Silva, J.V., Descheemaeker, K., Van de Ven, G., Schut, A.G.T., van Wijk, M. et al. 2021. The future of farming: who will produce our food? *Food Security*, 13: 1073-1099. <https://doi.org/10.1007/s12571-021-01184-6>

climatic conditions (Benin, 2016). This technology gap⁵ results in persistently low input adoption rates, leaving farming systems ill-equipped to withstand climate shocks – a particular vulnerability given the complexity of African agroecological systems and the inadequacy of mitigation and adaptation tools (Nyambo et al., 2022; Benin, 2016; Jayne and Sanchez, 2021).

Labour productivity remains constrained by insufficient development of labour-saving technologies, limited profitable employment alternatives beyond agriculture, limited income diversification opportunities, and a low-skilled workforce (Benin, 2016). These challenges are compounded by inadequate government support and weak private sector engagement – both of which hinder agricultural mechanization – as well as underdeveloped financial systems that restrict access to credit and insurance. Infrastructure deficiencies, gender inequality, and health challenges are also major factors (Benin, 2016).

Adding to the challenges to productivity growth is the problem of small and declining farm sizes. Even substantial yield improvements often translate into only modest income gains for smallholder producers, discouraging investment in modern technologies and improved farming practices (Giller and Andersson, 2024) – and perpetuating the low productivity levels the improvements were meant to address.

⁵ This adds to the challenge of mechanization – particularly the lack of affordable, small-scale solutions adapted to smallholder farmers – where high costs and limited access to appropriate technologies remain other major barriers.

Soil health is a key determinant of Africa's agricultural productivity and food security, shaping both crop yields and nutritional quality (Montgomery *et al.*, 2022; Montgomery and Biklé, 2021). Poor soil health remains the major biophysical constraint on agricultural production in SSA (Titttonell and Giller, 2013), and its consequences extend well beyond agronomy. Nutrient-deficient soils produce nutrient-deficient crops, contributing to micronutrient deficiencies in human diets – commonly called “hidden hunger”. The World Health Organization (WHO) estimates this affects more than two billion people globally, with disproportionate impact in Africa, where soil and human zinc deficiencies often overlap (Cakmak, McLaughlin and White, 2017). Practices that prioritize short-term yields ahead of soil health have deepened the problem (Mayer *et al.*, 2020). Restoring soil health is therefore essential not only for raising agricultural productivity but for improving nutrition and public health outcomes across the continent (FAO, 2022a).

Despite widespread awareness of the importance of healthy soils, degradation⁶ is serious, and accelerating. The Food and Agriculture Organization of the United Nations (FAO) estimates that one-third of soils globally are already degraded (FAO, 2021a), which could increase to 90 percent by 2050 (FAO, 2022b). **Africa's situation is more acute still: up to two-thirds of productive land (65 percent) is currently degraded** (Mansourian and Berrahmouni, 2021).

Ineffective land management is a primary driver of this degradation. Although smallholder farmers are the principal stewards of Africa's agricultural land, their ability to invest in sustainable practices is constrained by limited resources, insecure land tenure, weak incentives, and ongoing land fragmentation driven by inheritance customs and laws. Small farm sizes and declining soil health compound one another in what Giller and Andersson call a “double poverty trap” (Giller and Andersson, 2024) – where poor soils reduce yields and low yields leave farmers without the means to restore soil health.

ISFM offers a pathway forward, combining the judicious use of mineral fertilizers with bio-based inputs, improved germplasm, and locally adapted agronomic practices. However, the potential of this approach is held back by the prevalence of both excessive and insufficient fertilizer application, which continues to impede soil fertility regeneration (Vanlauwe *et al.*, 2023).

Attitudes and approaches to soil fertility have evolved considerably, as summarized in Table 1. Vanlauwe *et al.* (2023) trace the shifts: from the green revolution's emphasis on high-yielding varieties and synthetic fertilizers, to the emphasis on smallholder systems, biological processes and low-external-input sustainable agriculture (LEISA) of the 1990s. ISFM subsequently emerged as a pragmatic approach that emphasized optimizing the combined use of mineral and bio-based nutrients and of fertilizers as an entry point into the sustainable intensification of smallholder systems. Concurrently, concerns about soil erosion and loss drove major investments in conservation agriculture. More recently, agroecology and regenerative agriculture have gained attention, both emphasizing soil health as central to productivity growth and ecosystem services. Agroecology and regenerative agriculture are often interpreted as emphasizing minimal use of fertilizer and other inputs. Vanlauwe *et al.* (2023) argue that across all these approaches, environmental, social and human dimensions must be integrated into efforts to build and sustain soil health.

6 Soil degradation is defined as a change in the soil health status resulting in a diminished capacity of the ecosystem to provide goods and services for its beneficiaries. Degraded soils have a health status such that they do not provide the normal goods and services of the particular soil in its ecosystem. (See FAO, 2021a).

TABLE 1

Evolution of soil fertility management paradigms

TIME PERIOD	PARADIGM/ APPROACH	KEY CHARACTERISTICS	PRIMARY FOCUS	INPUT STRATEGY	TARGET AUDIENCE	LIMITATIONS/ CHALLENGES
1960s-1980s	First soil management paradigm (green revolution)	- High-yielding crop varieties - Use of mechanization - Irrigation systems - Fertilizer-intensive approach	Meeting nutrient demands of new crop varieties	Heavy reliance on mineral fertilizers	Farms with sufficient assets and resources	- Increased labour for weed control - Difficulty maintaining mulch cover - Limited smallholder adoption
1980s-1990s	Second soil management paradigm	- Focus on smallholder farms - Biological processes emphasis - Improved fallows (leguminous) - Bio-based input integration	Optimizing nutrient cycling and minimizing external inputs	Combined use of mineral fertilizers and bio-based inputs	Smallholder farmers on marginal lands	- Labour-intensive practices - Limited immediate benefits - Insufficient bio-based resources available - Poor farmer adoption rates
1980s-1990s	Low-external-input sustainable agriculture (LEISA)	- Minimal external inputs - Sustainable farming practices - Environmental focus	Sustainability with reduced input dependence	Minimal external inputs, focus on biological processes	Smallholder farmers	- Low yields insufficient for food security - Cannot address poverty effectively - Limited by nutrient deficiencies
2000s-present	Integrated soil fertility management (ISFM)	- Stepwise approach - Fertilizer as entry point - Efficient use of all resources - Soil rehabilitation focus	Combining all bio-based and mineral resources efficiently	Start with mineral fertilizers, then incorporate bio-based resources	Smallholder farmers	- Requires initial investment in fertilizers - Context-specific implementation - Depends on local resource availability
2000s-present	Conservation agriculture	- Three pillars: minimal tillage, soil cover, crop diversification - Fourth principle: appropriate fertilizer use - Soil erosion prevention	Soil health through conservation practices	Mineral fertilizers combined with conservation practices	Both large-scale and smallholder farmers	- Increased labour for weed control - Difficulty maintaining mulch cover - Limited smallholder adoption - Mainly successful on larger mechanized farms
Recent (third paradigm)	Agroecology	10–13 principles including diversity, synergies, recycling - Holistic ecosystem approach - Multiple ecosystem services	Ecosystem services beyond crop productivity	Variable interpretation, often minimal external inputs	Diverse farming systems	- Often misinterpreted as anti-fertilizer - May not address nutrient scarcity - Contradictory principles in some interpretations
Regenerative agriculture	Recent	- Water infiltration and erosion prevention - Building soil carbon - Plant diversity - Livestock integration - Reduced external inputs	Positive environmental impact of agriculture	Emphasis on biological processes with reduced external inputs	Various scales of farming	- Somewhat nebulous concept - Contradictory principles (e.g. no-till + no herbicides) - May not address immediate nutrient needs - Often misinterpreted to exclude fertilizers

SOURCE: Vanlauwe, B., Amede, T., Bationo, A., Bindraban, P., Breman, H., Cardinael, R., Couédel, A. et al. 2023. *Fertilizer and soil health in Africa: the role of fertilizer in building soil health to sustain farming and address climate change*. Muscle Shoals, IFDC. <https://ifdc.org/wp-content/uploads/2023/02/Fertilizer-and-Soil-Health-in-Africa-The-Role-of-Fertilizer-in-Building-Soil-Health-to-Sustain-Farming-and-Address-Climate-Change.pdf>

This report addresses the role of soil health in sustainable land productivity growth as a foundation for improved food security in sub-Saharan Africa.

Political recognition of the importance of soil health and fertility has grown, as reflected in the Nairobi Declaration of the 2024 Africa Fertilizer and Soil Health Summit. African leaders affirmed that building soil health and regenerating degraded soils are critical to sustainable agrifood systems transformation – requiring ISFM, efficient use of mineral and bio-based fertilizers, and sustainable soil and water management practices.

1.2 OBJECTIVE

This report addresses the role of soil health in sustainable land productivity growth as a foundation for improved food security in sub-Saharan Africa. It emphasizes soil fertility management approaches – including fertilizer use – needed to support more productive and sustainable agrifood systems, and the complementary need to improve farmers’ incentives to invest in soil health and productivity. Soil fertility management and fertilizer use have generated a vast literature, both institutional and academic (FAO, 2022b; World Bank, 2022b).⁷ Yet efforts to raise productivity and improve fertilizer use in SSA have achieved only limited success.

A new approach is therefore needed. Sustainable intensification will be central to delivering economic, social, and environmental gains for smallholder farms and the region as a whole. Success will require nutrient management that increases crop productivity while ensuring efficient use of nutrients at the higher fertilizer application rates of intensification models (Vanlauwe *et al.*, 2023). It will also depend on direct measures to address the economic constraints smallholder farmers face.

This report explores actions to mitigate further soil degradation and rehabilitate those already degraded, and to incentivize fertilizer use that balances long-term soil health and agricultural productivity with the immediate financial and nutritional needs of smallholder farmers.

⁷ For a broad overview of the literature, see FAO (2022b), World Bank (2022b), AFFM (2022), ECOWAS and World Bank (2023), and AGRA (2019), as well as Van Rooyen, Bjornlund and Pittock (2021), Bonilla Cedrez, Chamberlin and Hijmans (2021), and Falconnier *et al.* (2023).



1.3

REPORT STRUCTURE

This report is organized as follows. Section 1 describes the persistent food insecurity and low agricultural productivity in sub-Saharan Africa, highlighting the structural, environmental, and economic constraints that hinder sustainable growth, the critical role of soil health in improving yields and nutrition, and the evolution of soil fertility management approaches. Section 2 provides an overview of soil patterns, soil health and soil degradation in sub-Saharan Africa. Section 3 examines the structural, agronomic, and economic challenges affecting fertilizer use and efficiency. Section 4 explores the growing awareness of and opportunities to improve soil health, including the political economy of fertilizer subsidies and developments in policies and programmes over the past 20 years. Section 5 sets out policy and investment options with scope for the next generation of agrifood system policies. Section 6 brings together the conclusions.



The soil challenges in sub-Saharan Africa

2.1 A DIVERSITY OF SOIL TYPES AND QUALITY

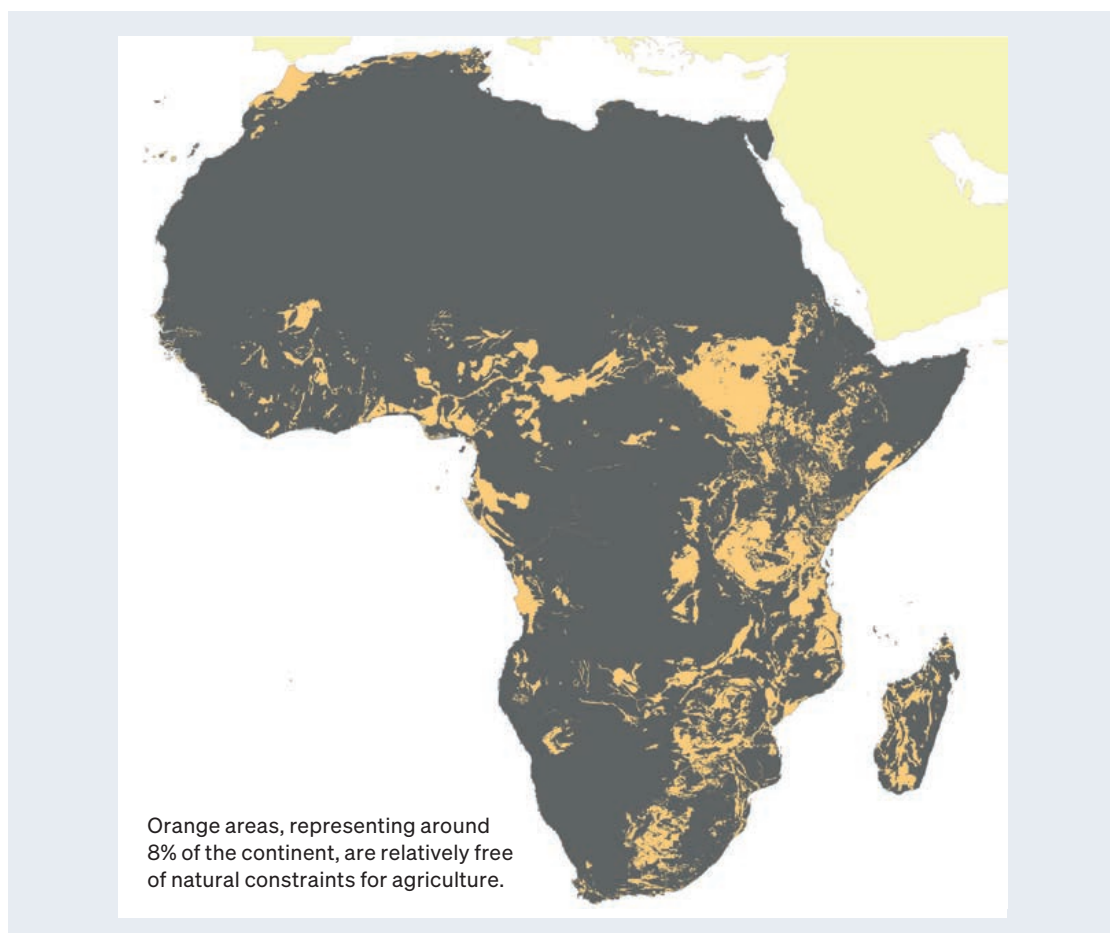
Africa spans an exceptional range of climates and geology, and the continent's soils reflect this diversity (Jones et al., 2013). Weakly developed soils such as Arenosols dominate the Sahara, Kalahari, Namib, and the Horn of Africa, encircling a central core of tropical soils. Ferralsols – strongly weathered and leached soils associated with high rainfall and old land surfaces – are widespread in Central, Eastern and Southern Africa. Gleysols, characterized by seasonal waterlogging, dominate extensive wetlands such as along the Niger, Congo, and Zambezi rivers. The Nile Valley, meanwhile, features vast stretches of heavy clay soils classified as Vertisols (Jones et al., 2013), which have high shrink–swell capacity and pose distinctive management challenges (IUSS Working Group WRB, 2022).

More than half of African land is unsuitable for rainfed agriculture.

This 55 percent of the continent's surface encompasses arid regions, salt pans, sand dunes, rocky terrain, and steeply sloping land. These regions sustain approximately 250 million inhabitants – representing 25 percent of the continental population in 2014 – predominantly through nomadic pastoralism (Jones et al., 2013). Some of these areas could potentially be brought into cultivation through irrigation.

Approximately 30 percent of African land, equivalent to 9 million km², has arable potential. Scarcely 16 percent of this area is high-quality soil, and 13 percent medium-grade, relatively free of major soil constraints to agriculture. Together, these categories represent approximately 8 percent of Africa's land mass. These areas support approximately 400 million people, or 40 percent of

FIGURE 3
Soil constraints for agriculture in Africa



SOURCE: Jones, A., Breuning-Madsen, H., Brossard, M., Dampha, A., Deckers, J., Dewitte, O., Gallali, T. et al., eds. 2013. *Soil Atlas of Africa*. Luxembourg, European Commission. <https://doi.org/10.2788/52319>

Africa's population, and benefit from consistent precipitation patterns and soils largely free of nutrient deficiency or insufficient depth. They are suited to uses ranging from crop cultivation, plantation agriculture, and forestry to aquaculture (Jones et al., 2013). The distribution of these fertile areas varies considerably across the continent, resulting in significant regional disparities in agricultural potential (see Figure 3) (Jones et al., 2013).

Soil quality assessment extends beyond general fertility metrics and should be evaluated in the context of specific crop requirements. Successful maize cultivation, for instance, requires the consideration of seven critical soil parameters: nutrient availability, nutrient retention capacity, root development conditions, soil aeration, salinity levels, absence of phytotoxic elements, and soil workability (Jones et al., 2013). Yet SSA lacks detailed soil nutrient maps at the microscale (Hengl et al., 2017) – essential for crop planning based on soil characteristics – which significantly limits the precision of agronomic guidance across the region.

Soil health challenges in SSA are compounded by farming practices that deplete soil nutrients at a rate far exceeding their replenishment, particularly in monoculture cereal systems where little crop residue remains in the field after harvest. A 1995 study estimated nutrient losses over 25 years at approximately 700 kg of nitrogen, 100 kg of phosphorus, and 450 kg of potassium per hectare across 100 million hectares of cultivated land (Sanchez and Leakey, 1997), with average N depletion rates reaching 22 kg/ha annually between 1970 and 2000 in 37 African countries (Sanchez, 2002). In financial terms, the losses are stark: SSA loses approximately USD 4 billion worth of soil nutrients annually through nutrient mining (Vanlauwe *et al.*, 2023), making it the only region with persistently negative nutrient balances (Vanlauwe *et al.*, 2023). This chronic depletion has direct consequences in declining per capita food production on smallholder farms (Hengl *et al.*, 2017).

Macronutrient losses vary considerably depending on the chemical properties of each nutrient and its interactions with soil, water and wind. (For detailed definitions and measures, see Section 3.5 and Annex 1.)

Farmers' ability to replenish soil nutrients is constrained by a web of interacting factors: the long time before investments in soil health pay off; land tenure insecurity; liquidity and credit constraints (Vanlauwe *et al.*, 2023); the high cost of mineral fertilizers; and limited access to farm machinery. Demographic pressure on land compounds these factors, making traditional restorative practices such as long fallow periods increasingly difficult to maintain (Giller, 2020). Faced with these constraints, many farmers turn to land expansion – often involving deforestation – rather than fallowing, seeking short-term gains from temporarily more fertile soils. This search for fertility through agricultural expansion accounts for roughly 90 percent of current global deforestation (FAO *et al.*, 2024).

Continued low investment in soil fertility cannot be explained by market failures or information gaps alone – it also reflects the underlying heterogeneity of smallholder farming systems. Differences in plot size, market access, and livelihood orientation fundamentally shape both soil management decisions and productivity outcomes. Subsistence farmers cultivating very small plots face a distinct set of constraints: investments in soil fertility compete directly with immediate food security needs, while uncertain returns, climate risk, poor markets, and limited access to labour and credit amplify the cost of failure. This is borne out empirically – nutrient balances are consistently negative on farms where large areas are devoted to subsistence cereal production (Nkonya *et al.*, 2005), while positive balances appear on mixed farms where farmers apply manure or cultivate high-value cash crops (Onduru *et al.*, 2007; Vanlauwe *et al.*, 2015). Farm typology studies across SSA consistently identify patterns of variation linked to plot size, resource endowment, and production objectives, confirming that farmer type – rather than knowledge alone – determines soil management outcomes (Chikowo *et al.*, 2014). In this context, low fertilizer use often reflects a rational response to risk rather than a lack of knowledge or motivation – a distinction that is central to any accurate diagnosis of soil fertility decline (Dercon and Christiaensen, 2011; Tittonell and Giller, 2013).

Fertilizer non-responsiveness adds another layer of complexity. A recent study on maize and soybean found non-responsiveness ranging from 0 percent to 68 percent across sites and seasons, varying by crop and field conditions such as irregular rainfall, soil texture, and nutrient imbalances. Soybean was most affected by low phosphorus and high silt content, while non-responsiveness in maize was linked to high silt and cation imbalances (Roobroeck et al., 2021). This wide variability underlines the limits of uniform fertilizer recommendations and reinforces the case for more targeted, site-specific approaches.

Addressing these challenges thus requires approaches explicitly grounded in the diversity of smallholder farming systems, rather than blanket recommendations that assume homogeneity. Central to this are strategies such as intercropping, crop rotation, and crop–livestock integration, not as transitional steps toward input-intensive systems, but as core strategies for maintaining soil health and managing production risk in the absence of external inputs. These practices shape nutrient cycling, organic matter accumulation, and fertilizer responsiveness in ways that differ fundamentally from monoculture or continuous monocropping, which drive patterns of nutrient depletion and declining organic matter. Effective targeting requires moving from regional-scale recommendations to farm- and field-scale approaches, matched to site-specific biophysical conditions, available resources, and household objectives (Vanlauwe et al., 2014).

Cropping system choice is also closely bound up with farm size. Very small plots often constrain the feasibility of crop rotation or fallowing, pushing farmers toward repeated cultivation of the same crop and accelerating soil degradation. Larger or more commercially oriented farms are better positioned to diversify production, spread risk, and adopt soil-restoring practices whose benefits materialize over time. Ultimately, soil health challenges are inseparable from the production systems and land constraints that shape what farmers can realistically do.

Taken together, this evidence suggests that soil fertility is best understood as a conditional constraint: its importance for productivity growth is undeniable, but the degree to which it actually limits productivity varies considerably depending on the farmer and the context. For smallholder subsistence farmers operating under tight economic and ecological constraints, easing soil fertility bottlenecks requires coordinated responses that address risk, incentives, and complementary investments alongside agronomic considerations.

2.2 SOIL DEGRADATION UNDERMINES AGRICULTURAL LAND PRODUCTIVITY AND FOOD SECURITY IN SSA

Soil degradation – the alteration of its physical, chemical, and biological properties in ways that reduce its fertility and productive capacity – is a pervasive problem across Africa, though estimates of the extent vary considerably. FAO and ITPS (2015a) estimate that close to 500 million hectares of land in SSA are affected, representing more than 20 percent of land in most countries in the region.⁸ Other assessments put the share of degraded productive land at 65 percent (Mansourian and Berrahmouni, 2021), or even as much as 75–80 percent of the continent's cultivated area (AUDA-NEPAD, 2023), with annual associated nutrient losses of 30 kg–60 kg of nutrients per hectare (AUDA-NEPAD, 2023). A further estimate suggests that 46 percent of African

⁸ This covers any change in soil health status (FAO and ITPS, 2015a).

lands are degraded, potentially affecting the livelihoods of 485 million people (AGNES, 2020). The divergence between these figures may reflect differences in methodology, scope and definitions, but all point to a problem of substantial and growing scale.

Soil degradation manifests through four main processes: erosion, organic matter depletion, nutrient depletion, and loss of soil biodiversity. Water and wind erosion account for approximately 84 percent of all soil degradation, with chemical degradation accounting for a further 12 percent and physical degradation for the remaining 4 percent (FAO and ITPS, 2015a). Erosion and physical degradation share many of the same drivers, including overgrazing, deforestation, and unsustainable cultivation practices – such as complete crop residue removal. Where soils are stripped of vegetation cover and organic matter, they become more vulnerable to water and wind erosion, less able to retain water and support plant growth, and less responsive to even appropriate fertilizer use. Chemical degradation is associated with unbalanced or absent fertilizer use (FAO, 2022a), as well as pollution from pesticide overuse or other contamination.

Expansion of agriculture into marginal areas and demographic pressures on existing farmland add further complexity (FAO, 2022a), contributing to land fragmentation and insecure tenure arrangements that discourage long-term soil management.

The consequences for productivity are severe. Yield losses on degraded agricultural land range from moderate – approximately 2 percent over several decades – to catastrophic, exceeding 50 percent in the worst-affected areas, depending on crop, soil type, climate and production system (FAO, 2022a).

Acidification, salinization, and desertification present additional, significant threats to soil health across SSA. Acid soils – covering up to half of the region – reduce health through aluminium toxicity and phosphorus fixation, while salinization driven by poor irrigation management and rising evaporation disrupts soil structure and nutrient uptake. Desertification accelerates nutrient loss and erosion, especially at the margins of arid and semi-arid zones, diminishing soil productivity and resilience. Together, these processes compound existing degradation, further threatening crop yields, biodiversity, and water regulation across the continent (Mesele *et al.*, 2025).

On degraded soils, nutrient deficiency cannot be addressed simply by increasing application of mineral fertilizer. Low levels of soil organic matter impede nutrient uptake by plants, while poor water retention means that much of any fertilizer applied is lost through leaching before the nutrients can be absorbed by crops. Rehabilitation must therefore precede efficient fertilizer use. This process typically unfolds over several years and requires sustained external inputs, including both mineral and bio-based fertilizers, soil amendments, and lime applications – alongside conservation practices. Approaches that rely on mineral fertilizer alone, without first addressing the underlying soil conditions, are unlikely to achieve durable improvements in soil health or yields (Vanlauwe *et al.*, 2023).



Fertilizer underused, often misused in SSA

3.1 AVERAGE FERTILIZER APPLICATION RATE WELL BELOW THE WORLD AVERAGE

Promoting fertilizer use is a central element of agricultural policy in many countries and remains a popular approach for increasing agrifood productivity (see Box 1) (Venkateswaran *et al.*, 2025). Many countries use subsidies to make fertilizers more affordable to producers, and these can represent substantial expenditures in national budgets (Zaveri, 2025). In Africa, nations allocate 14–26 percent of their agricultural budgets to fertilizer and input subsidies (Jayne *et al.*, 2018), with nitrogen constituting the bulk of those applied on African soils (Sheahan and Barrett, 2017). Under- or over-subsidization of specific fertilizer nutrient can lead to inefficient use (Sapkota and Bijay-Singh, 2025).

Considering N, P, and K use together (see Figure 4), fertilizer use in Africa has increased gradually since 1961, but the growth rate is dwarfed by the increase in use per hectare in Asia and South America. Average nitrogen fertilizer use rates in Africa are estimated at 13.3 kg/ha in 2022, compared to a global average of 65.4 kg/ha (Figure 5). Africa's nitrogen application per hectare increased in relative terms at a higher rate than the world average from 1961 (Figure 7), though from a lower base (FAO, 2021b). While Africa has a similar nitrogen/yield relationship as the global average, the much lower use of nitrogen implies much lower yields per hectare.

Fertilizer application in Africa varies widely, influenced by agronomic and economic conditions and value chain specificities across regions and crops. Among SSA countries, nitrogen fertilizer use ranged from 3.2 kg/ha in 2022 on average in Middle Africa⁹ to 37.2 kg/ha in Southern Africa.¹⁰ Figure 6 shows the variation in fertilizer consumption (kg/ha of arable land) across African countries, with darker shades indicating higher consumption levels (IndexMundi, 2019).¹¹ Proximity to ocean ports and resulting lower inland transportation costs

⁹ Angola, Cameroon, Central African Republic, Chad, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, and São Tomé and Príncipe.

¹⁰ Botswana, Eswatini, Lesotho, Namibia, and South Africa.

¹¹ Underlying data are drawn from FAOSTAT. IndexMundi. 2019. Fertilizer consumption (kilograms per hectare of arable land) – Africa. In: *Agricultural Production*.

BOX 1**The role of fertilizers in soil fertility and agricultural development**

Fertilizer refers to a chemical or natural substance or material used to provide nutrients to plants, usually via application to the soil, but also to foliage or through water by fertigation, hydroponics, or aquaponics operations (FAO, 2019). (For details of different fertilizer types see Annex 2). Crop production depends on the adequate availability of nutrients for plant growth and water to facilitate the conversion of atmospheric carbon dioxide (CO₂) into organic matter through photosynthesis. It requires various nutrients, including nitrogen (N), phosphorus (P), potassium (K), magnesium (Mg), sulphur (S), and micronutrients such as iron and zinc. Significant quantities of nutrients are removed from agricultural land when crops are harvested without replenishment (Falconnier *et al.*, 2023).

Fertilizer application has shaped agricultural development and land productivity increases globally for decades. It has been estimated that almost half the people in the world are fed because of the use of synthetic nitrogen fertilizer (Erisman *et al.*, 2008). The use of nitrogen as a fertilizer has doubled global rates of nitrogen fixation, resulting in increases in crop yields of between 30 percent and 50 percent and lifting billions of people out of food insecurity (Erisman *et al.*, 2008; Stewart *et al.*, 2005). Given the current estimates of soil depletion in SSA, combined with existing cropping systems, these essential nutrients cannot be adequately replenished without the application of fertilizers. When applied appropriately, they enhance soil fertility, compensate for nutrient depletion, and significantly increase crop yields (FAO, 2019).

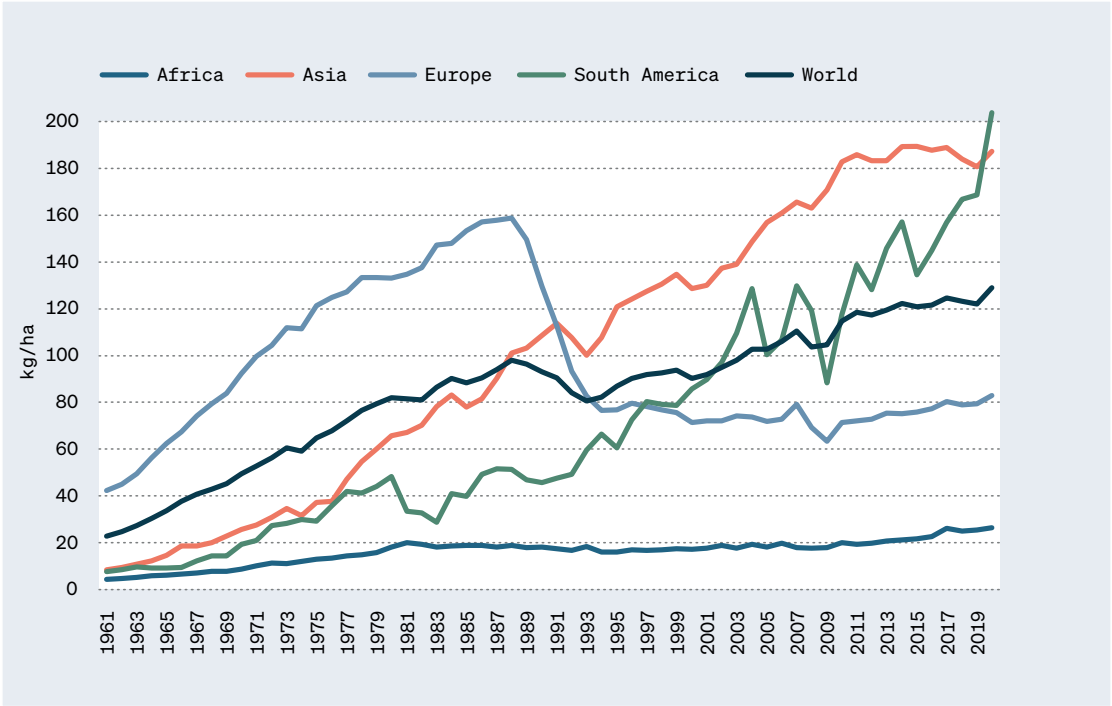
are critical factors affecting the cost and use of fertilizer. Fertilizers are used primarily in horticulture, cash crops and large commercial holdings, while in most countries smallholders use relatively little (Falconnier *et al.*, 2023; Vanlauwe *et al.*, 2023). Spatial variation in price and profitability are also factors, as discussed in Section 3.3.

The nitrogen and phosphorus challenge in Africa stands in contrast to global trends. Many parts of the world exceed safe levels of both nitrogen and phosphorus, according to planetary boundary studies (Steffen *et al.*, 2015; Richardson *et al.*, 2023), while most African agricultural systems operate within “safe” boundaries, reflecting an underuse of nitrogen fertilizers rather than effective management. From a global resource-efficiency perspective, optimal nitrogen distribution would increase fertilizer use in SSA – including mineral fertilizer – while reducing excessive application in other parts of the world (Gerten *et al.*, 2020), especially mineral fertilizer.

Despite already low consumption levels, recent years have witnessed declining fertilizer use trends across Africa. Between 2021 and 2022, agricultural fertilizer consumption declined across all major nutrients: nitrogen decreased by 9 percent, potassium by 6 percent, and phosphorus by 5 percent. These decreases can be attributed to high international fertilizer prices, currency devaluation, and logistical disruptions that further constrained access to essential nutrients (FAO, 2024b).

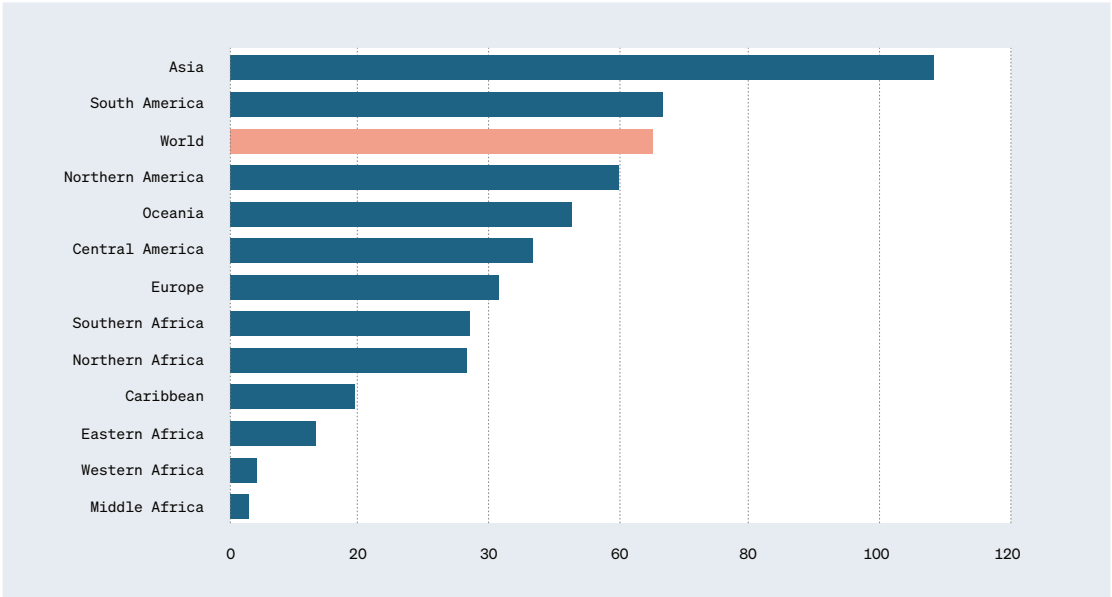
Charlotte, USA, IndexMundi. [Accessed on 6 May 2026]. www.indexmundi.com/facts/indicators/AG.CON.FERT.ZS/map/africa

FIGURE 4
Regional fertilizer use per area of cropland (sum of nitrogen, phosphate and potash), 1961-2020



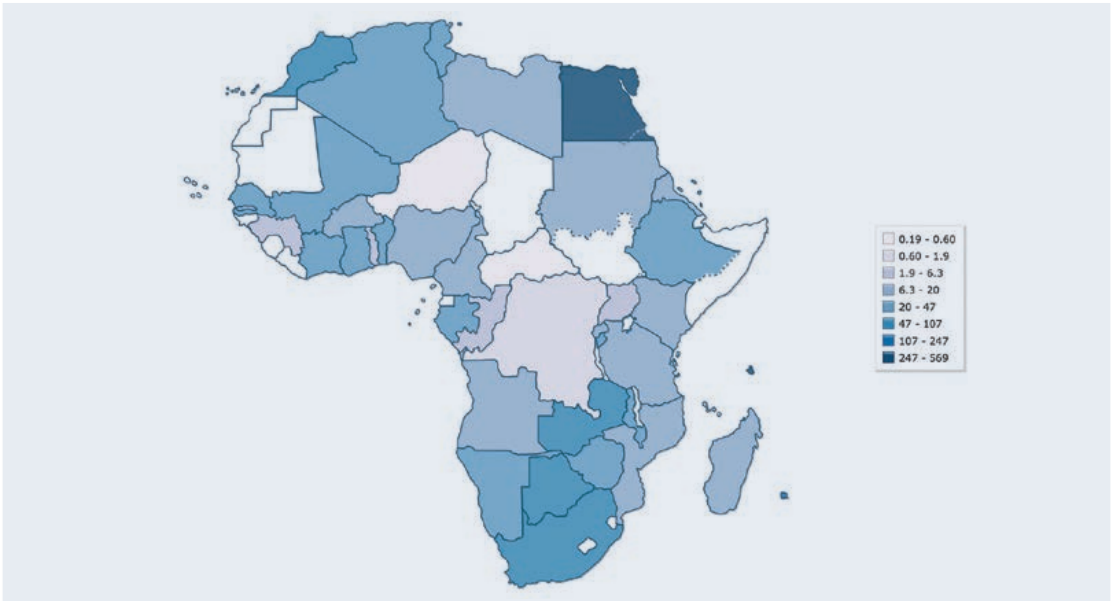
SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN. Licence: CC BY-4.0.

FIGURE 5
Use of nitrogen per area of crop land (kg/ha), 2022



SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN. Licence: CC BY-4.0.

FIGURE 6
Mineral fertilizer consumption in Africa (kg/ha of arable land)

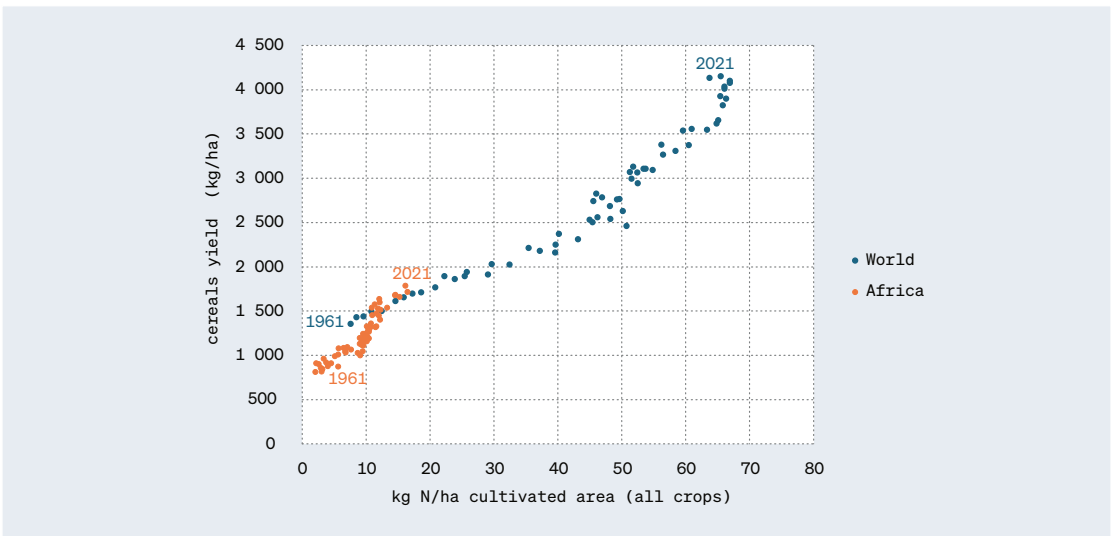


NOTE: Fertilizers here comprise nitrogenous, potash and phosphate fertilizers (including ground rock phosphate). Underlying data are drawn from FAOSTAT: Fertilizers by nutrient.

Refer to the disclaimer on page II for the names and boundaries used in this map. The final boundary between the Republic of Sudan and the Republic of South Sudan has not yet been determined. The final status of the Abyei area is not yet determined.

SOURCE: IndexMundi. 2019. Fertilizer consumption (kilograms per hectare of arable land) – Africa. In: Agricultural Production. Charlotte, USA, IndexMundi. [Accessed on 6 May 2026]. <https://www.indexmundi.com/facts/indicators/AG.CON.FERT.ZS/map/africa>

FIGURE 7
Cereal yield and nitrogen use per hectare, 1961-2021



SOURCES: Authors' own elaboration from: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. <https://www.fao.org/faostat/en/#data/RFN>. Licence: CC BY-4.0.; FAO. 2024. FAOSTAT: Crops and livestock products. [Accessed 29 April 2026]. <https://www.fao.org/faostat/en/#data/QCL>. Licence: CC BY-4.0.



3.2 COSTS, MARKET STRUCTURE AND MARKET FAILURES LIMIT FERTILIZER USE

The challenges of fertilizer use in sub-Saharan Africa are fundamentally rooted in high prices, market structure and market failures, which give rise to a series of interconnected supply and demand constraints examined in this section (Abate, Abay and Spielman, 2020; Dillon and Barrett, 2017; Vanlauwe *et al.*, 2023; Isik, 2002). Countries that rely heavily on fertilizer imports are particularly exposed to price volatility and supply disruptions – a vulnerability especially pronounced across much of Africa, where even moderate price increases can significantly reduce fertilizer use. African farmers who rely on fertilizers depend heavily on imports from producers in the Persian Gulf and may face significantly increased risks due to the closure of the Strait of Hormuz in 2026. A protracted conflict would have serious implications for input trade, agricultural production and food security across the continent (FAO, 2026a).

3.2.1 Mineral fertilizer prices well above the world average

The most immediate constraint is cost – fertilizers are expensive in sub-Saharan Africa. Most of those used in SSA are imported (see Section 3.3), and fertilizer prices are the highest in the world, up to four times higher than in Europe (Hernandez and Torero, 2018). This disparity stems primarily from high logistical costs, including international shipping to African ports, port handling, and inland distribution through inefficient networks. These supply challenges are further compounded by limited competition in a highly concentrated global fertilizer industry, which may contribute to persistently high prices in SSA. Further detail on price formation is provided in the following sections.

3.2.2 Extensive supply-side constraints

Numerous studies have highlighted supply-side constraints in fertilizer distribution as a key economic factor limiting fertilizer use in most African countries (Binswanger and Sillers, 1983; Dercon and Christiaensen, 2011; Karlan *et al.*, 2014). These constraints include high import prices and transport costs, contributing to high farm-gate prices and inconsistent availability (Hernandez *et al.*, 2018). Further supply-side challenges include inadequate infrastructure, limited market competition, inefficient distribution systems, and insufficient private sector investment. Evidence from the green revolution, particularly in India, demonstrates the importance of research and development, and of crucial

public investment in infrastructure and services – particularly roads, irrigation systems, rural electrification, and agricultural extension (Johnson, Hazell and Gulati, 2003) – as supply-side factors in agricultural productivity growth.

Distribution challenges are among the most significant of these constraints in SSA. The cost of transporting fertilizers to landlocked countries can nearly double the fertilizer price (IFDC, 2020; Nduva and Migwe, 2022). The expenses associated with ocean shipping¹² and inland transportation¹³ are higher than in other parts of the world and median trade costs are more than five times higher (Porteous, 2019). While fertilizer prices in SSA tend to increase with increasing distance from main markets, exceptions to this pattern in Nigeria, Ghana and Benin suggest other factors besides transaction costs may be at play, including relative demand, imports from neighbouring countries and subsidy policy (Bonilla Cedrez *et al.*, 2020).

Given that fertilizer production relies heavily on natural gas as a key input, energy shocks transmit directly into fertilizer markets. An oil price increase of 10 percent due to a geopolitical oil supply shock has been associated, on average, with increases in natural gas prices that have peaked at about 7 percent after 11 months, and increases in fertilizer prices that have peaked at a little over 5 percent after 12 months. The energy price shocks of early 2026 are thus significant leading indicators in this context, particularly given that these price increases have tended to show notable persistence (World Bank, 2026).

In West Africa, supply chain costs have been shown to add between 50 percent and 85 percent to the free on board (FOB) price before reaching farmers (Camara and Obanyi-Brobbe, 2021; USAID, Nitidae and IFDC, 2019). The baseline FOB price constitutes about 60 percent of the wholesale cost, with final prices varying substantially. For example, urea ranges from USD 319/tonne in Senegal to USD 475/tonne in Niger. Figure 8 provides a breakdown of the range of costs incurred in the imported fertilizer value chain. Key cost components include logistical expenses (20 percent), operational costs (12 percent), and financial costs such as duties and taxes (7 percent). Land transportation is particularly burdensome at USD 0.07/km for each tonne of fertilizer – more than double European rates around USD 0.03/km. Other factors affecting local prices include port efficiency, packaging costs, shipment size, and geographical location (Camara and Obanyi-Brobbe, 2021; USAID, Nitidae and IFDC, 2019).

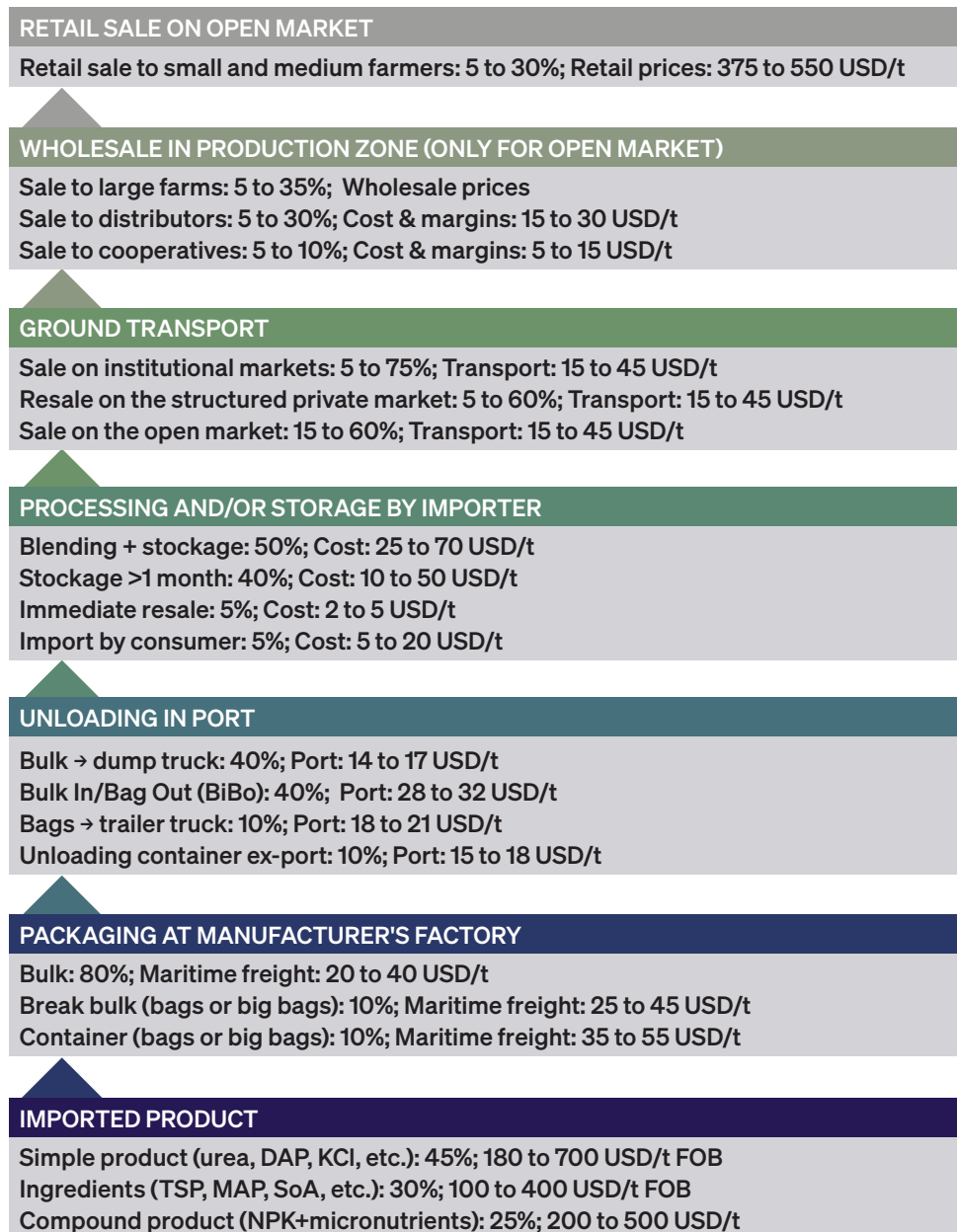
A study of four corridors in West Africa found that pricing structures vary considerably between countries, with prices typically higher in landlocked regions and areas distant from ports or distribution centres (USAID, Nitidae and IFDC, 2019). As mentioned in the context of Southern Africa, landlocked countries incur additional expenses in the form of direct and indirect transit costs. Direct costs include freight forwarding to inland countries along with associated customs duties and royalties; indirect costs encompass duties paid by importers in coastal countries for re-exporting fertilizer to dealers inland. The burden is significantly increased if the product requires unpackaging for blending at

¹² For seaborne shipments, limited port capacity may force importers to use smaller vessels, increasing landed costs. Additionally, poor port infrastructure – aged wharveside equipment, limited warehousing and inadequate handling capacity – slows cargo discharge, increasing congestion and the risk of demurrage costs.

¹³ On land, limited rail infrastructure and poor roads impede the movement of cargo to inland countries. Of substantial burden too are the checkpoints truckers encounter on inland routes, where they may be obliged to meet demands for government fees or other payments in order to proceed.

FIGURE 8

Marketing process of imported fertilizers to coastal countries in West Africa with the volumes (percent) and cost range (USD/tonne) of each option at each stage



SOURCES: Adapted from Camara, O.M. & Obanyi-Brobbe, S. 2021. Fertilizer logistics in West Africa. *Fertilizer Focus*, 38(1). [Cited 8 April 2024]. <https://ifdc.org/2021/02/26/fertilizer-logistics-in-west-africa/>; USAID, Nitidae & IFDC. 2019. *Structure of logistics costs and fertilizer import procedures along 4 corridors in West Africa. Feed the Future Enhancing Growth through Regional Agricultural Input Systems (EnGRAIS)*. [Cited 2 May 2026]. <https://ifdc.org/wp-content/uploads/2020/09/IFDC-EnGRAIS-Cost-build-up-4-corridors-West-Africa-Aug-2019.pdf>

processing facilities and subsequent repackaging (Ornella and Fan, 2024).¹⁴ Prices also vary significantly within countries. A study across 18 countries from 2010–2018 revealed significant internal price variations compared to national averages (Bonilla Cedrez et al., 2020).

3.2.3 Significant demand-side limitations

Fertilizer use is also constrained by several demand-side factors, including inadequate information on soil types and fertilizer recommendations, farmer perception of fertilizer quality, inadequate packaging sizes, low returns on investment, and limited access to agricultural extension services.

Farmers use little fertilizer when returns are low – when crop prices are too low relative to fertilizer costs, or when production risks are too high. Farmers in remote areas may not be able to sell surplus production profitably because local consumers have limited purchasing power and nearby markets are small. This problem persists even when people in these areas face hunger and food insecurity, because poor roads, high transport costs, and weak market connections prevent food needs from translating into actual market demand that farmers can respond to.

Figure 9 illustrates an important pattern: countries with larger non-agricultural sectors use significantly more fertilizer per hectare. Since a larger non-agricultural sector generates growing urban demand for food, this suggests that urbanization and economic diversification – not agriculture itself – help to drive fertilizer demand. Simply put, farmers invest in fertilizers when it is profitable, and reliable markets and a strong non-farm economy play important roles in making that possible.

Niger provides an example of this dynamic. With 83 percent of its population living in rural areas – one of the highest agricultural dependency rates in the world – Niger uses only 3 kg of fertilizer per hectare, among the lowest rates anywhere. Without sufficient urban demand and economic diversification, farmers lack the market incentives to invest in productivity-enhancing inputs – a dynamic in which functional input, credit and output markets play a major role.

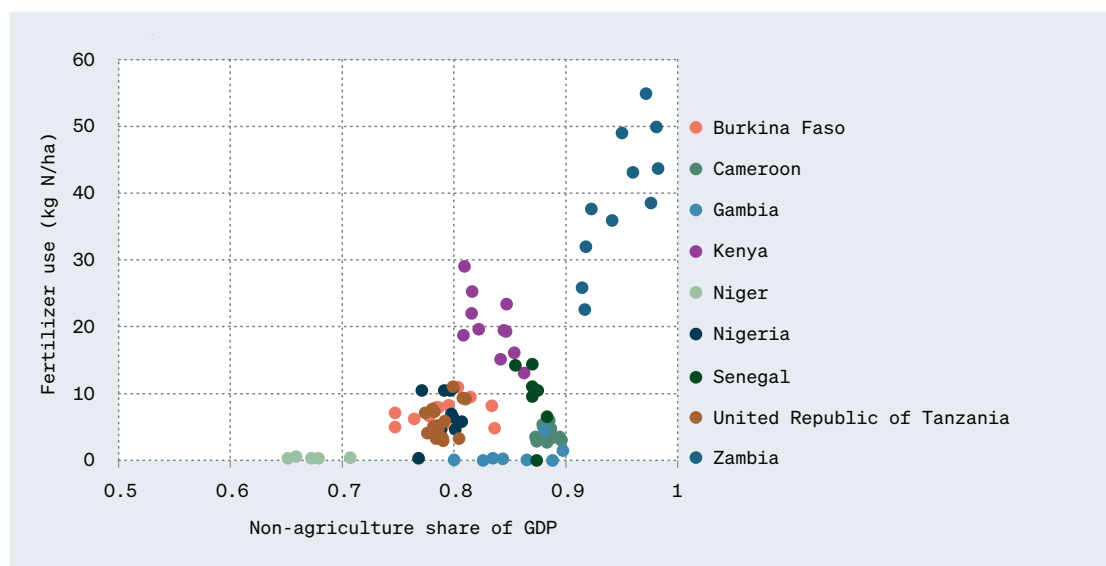
Several studies show that the considerable variations in returns on fertilizer use in Africa can be explained in part by soil type. There is robust evidence that low soil fertility and soil acidity can render mineral fertilizers unprofitable (Marenya and Barrett, 2009; Burke, Jayne and Black, 2017). Abate, Abay and Spielman (2020) note that different types and brands of chemical fertilizers contain varying types and levels of nutrients, implying that their profitability depends on specific soil nutrient requirements and application conditions. In most cases, soil testing and associated agronomic assessments are usually costly and inaccessible to smallholder farmers across most of the continent, and the spatial resolution of soil tests remains quite low (Carletto et al., 2017). Consequently, farmers may rely on inaccurate proxies for their soil properties (Berazneva et al., 2018).

Innovations in soil mapping and testing technologies offer potential solutions to the lack of soil property information, though these are still not widely

¹⁴ Africa has expanded its fertilizer-blending capacity significantly, with these facilities using pre-manufactured components to create customized products for local markets. Currently, 59 blending plants operate across sub-Saharan Africa, led by Nigeria (14), Côte d'Ivoire (6), and Ethiopia (5), with the remainder distributed across 11 other countries. These plants are strategically positioned near agricultural regions, and an additional 19 facilities are planned, including five more in Nigeria and four in the United Republic of Tanzania.

FIGURE 9

Correlation between countries, and within countries, of non-farm sector growth and fertilizer use



NOTES: Each dot represents a country for a given year, from 2005 to 2023. Figure elaborated using data drawn from four FAOSTAT datasets: Fertilizers by nutrient; Land use – arable land; Value added – agriculture, forestry and fishing; and Gross domestic product.

SOURCE: Authors' own elaboration from FAO. 2026c. FAOSTAT: Data. [Accessed 29 May 2026]. <https://www.fao.org/faostat/en>. Licence: CC-BY-4.0.

available at scale (Harou *et al.*, 2022; Murphy *et al.*, 2020; Ayalew, Chamberlin and Newman, 2022). Recent initiatives such as iSDAsoil provide gridded soil property predictions at 30-metre resolution across sub-Saharan Africa, representing the first continent-wide field-level soil maps with more than 20 soil properties predicted (iSDA, 2024). These maps use machine learning to analyse vast datasets from soil samples, satellite imagery, and geospatial analysis, offering unprecedented granularity compared to earlier 250-metre resolution maps (IFDC, 2024). In West Africa and the Sahel, updated digital soil maps have been designed to deliver localized agronomic recommendations through platforms such as AgWise (IFDC, 2024). By providing precise information on soil nutrient deficiencies and characteristics, these digital tools can enable farmers to make informed decisions about crop selection, planting schedules, and fertilizer application – potentially increasing yields by up to 40 percent while reducing fertilizer overuse (IFDC, 2024). Despite these technological advances, challenges remain in ensuring high-quality data collection, building digital infrastructure, and scaling access to smallholder farmers across the continent.

Poor fertilizer quality in Africa poses a significant challenge, as evidenced by studies in West Africa and Uganda, with key issues including nutrient deficiencies, short weight, and adulteration. Research in West Africa found that blended fertilizers frequently have severe quality issues, often failing to meet ECOWAS nutrient content standards due to insufficient nutrient input or granule segregation during manufacturing (Sanabria, Dimithe and Alognikou, 2013). In Uganda, nutrient shortages are a particular concern for liquid fertilizers, and deficiencies in granulated products are believed to originate during manufacturing (Sanabria *et al.*, 2018). In both studies, short weight is a problem, often



originating from manufacturing or importing facilities. While widespread adulteration was not confirmed in Uganda's larger bags, concerns remain about fillers being added when fertilizers are repacked into smaller bags for small-scale farmers (Sanabria *et al.*, 2018). Poor physical attributes such as caking and fine particles from manual handling and inadequate bagging can also degrade quality (Sanabria *et al.*, 2018; Sanabria, Dimithe and Alognikou, 2013).

Laboratory capacity for fertilizer quality assurance in SSA remains inadequate. Many countries lack accredited laboratories and standardized quality assurance and quality control systems aligned with international standards. Existing facilities are usually underequipped with limited capacity to perform nutrient analysis beyond NPK and suffer from poor maintenance and calibration practices. Infrastructure constraints such as outdated equipment, unreliable energy supply, and insufficient storage, all further compromise the analytical accuracy (Keyser *et al.*, 2015). As a result, inconsistencies in fertilizer quality persist across markets, undermining farmer confidence, reducing the effectiveness of fertilizer use, and limiting demand.

These challenges are often compounded by a lack of personnel with sufficient training in fertilizer quality control, fragmented regulatory frameworks, low levels of inspection, and limited enforcement of existing laws. Addressing these issues requires strengthening regulatory capacity across the supply chain, implementing pre-export verification systems and improving storage and handling practices among agrodealers (Sanabria *et al.*, 2018).

A World Bank review published around the time of the fertilizer price peak in mid-2022 underlined challenges in the physical quality of fertilizers, such as granule integrity and packaging. While these issues do not appear to reduce fertilizer efficacy, they can make application more difficult and negatively influence farmer perceptions of quality. The review noted that high and rising fertilizer prices – which by early 2022 had already surpassed the peaks of the 2007–2008 food price crisis – could alter the incentives for adulteration and counterfeiting. This suggests that fertilizer quality is dynamic and responds to market conditions. It is therefore crucial to continuously collect data on both fertilizer quality and farmer beliefs, as these may also change in response to price fluctuations and perceived shortages (Michelson, Gourlay and Wollburg, 2022).

Smallholder farmers' negative experiences with and perceptions of fertilizer quality may constitute another constraint to adoption, creating a complex dynamic that affects farmers' willingness to invest in agricultural inputs (Michelson *et al.*, 2021). Farmers rely on multiple observable signals to assess quality, including the physical appearance of the fertilizer, packaging condition, retail location, brand reputation, and observed crop yields. These perceptions are often reinforced by media coverage of adulteration cases, further complicating efforts to build trust in the market. Evidence from the United Republic of Tanzania demonstrates a disconnect between actual fertilizer quality and farmers' perceptions. While available fertilizers typically meet national quality standards, farmers maintain significant scepticism about product authenticity (Michelson *et al.*, 2021). This perception gap is reflected in purchasing behaviour: when farmers receive credible information about fertilizer quality, particularly regarding unobservable characteristics such as nitrogen content, their willingness to pay increases by approximately 50 percent. Without such information, farmers remain widely suspicious of the agronomic quality of the fertilizer in the market, suggesting a substantial information asymmetry (Michelson *et al.*, 2021).

Poor storage conditions and potential adulteration during repackaging may also discourage farmers from investing in fertilizers. To protect consumers from adulterated products and reduce quality control challenges and the risk of corruption (Chianu, Chianu and Mairura, 2012a), many sub-Saharan African countries prohibit fertilizer repackaging. However, packaging size presents an additional barrier to access and adoption for smaller-scale agricultural operations (Chianu, Chianu and Mairura, 2012a). Standard 50 kg bags, mandated by regulation or entrenched by commercial habit, are often too large and too costly for smallholder farmers. Smaller packages could better meet the time-based needs of such resource-constrained farmers, though they come at a higher cost per unit of weight.

Farmers' fertilizer use may also be impeded by informational or knowledge barriers. Ineffective agricultural extension services, for example, may hinder knowledge dissemination and, consequently, the adoption of improved technology. Farmers' organizations can serve as one mechanism for delivering agricultural extension services by promoting use of fertilizers, high-quality seeds, and yield-enhancing practices. Moreover, they have the potential to contribute to education, innovation, and market access. Such organizations may rely on support from governments and other agencies, however, and may lack the technical, managerial or leadership capacity to address these roles effectively. They may also be ill-equipped to support farmers constrained by infrastructure challenges (e.g. lack of irrigation, storage facilities, and feeder roads) or limited financial access (Bizikova *et al.*, 2020).

Beyond knowledge and organizational constraints, financial flows and timing further limit fertilizer adoption. A farmer's cash flow from agricultural income does not often coincide with input expenditure cycles, making it difficult to purchase fertilizer at the appropriate time. Farmers typically sell their crops at harvest time when market prices are lowest yet must purchase fertilizer at the start of the planting season when they may face credit and liquidity constraints. This mismatch particularly affects smallholder farmers. Evidence from four countries – Malawi, Nigeria, Uganda, and the United Republic of Tanzania – shows that credit use for fertilizer, pesticide or seed purchases is extremely low, with only 6 percent of farmers on average using any form of credit to buy inputs. Modern inputs are primarily financed through cash from non-farm activities and crop sales (Adjognon, Liverpool-Tasie and Reardon, 2017).

Moreover, fertilizers are often perceived as risk-increasing inputs: while they can enhance average crop yields, they also amplify yield variability under changing conditions such as rainfall and pest prevalence (Finger, 2012; Rajsic, Weersink and Gandorfer, 2009). As discussed in Section 2.1, this risk dynamic varies considerably across farm types. Semi-commercial and commercial farmers operating in well-connected markets face a fundamentally different risk calculus than subsistence smallholders, for whom fertilizer investment often competes directly with immediate food security needs and where the absence or inadequate performance of input, credit and output markets and extension services makes the cost of failure prohibitively high (Chianu, Chianu and Mairura, 2012a; Bizikova et al., 2020). Cash crop value chains such as cocoa, coffee and cotton offer models of contractual arrangements to ease smallholder liquidity constraints for the purchase of fertilizer and other inputs, often on credit (Tsakok and Guedegbe, 2019; Tefft, 2010).

Beyond soil properties, returns to fertilizer investment vary systematically with farm size, market connectivity, and agroecological endowment. Where fertilizer has delivered sustained productivity gains – such as in parts of Kenya, Ethiopia, and Nigeria – positive outcomes have typically been concentrated in better-endowed zones or among farmers with reliable market access and adequate infrastructure and services (e.g. roads, extension and credit). In Nigeria, fertilizer use and yield improvements have been markedly higher in well-connected states than in regions lacking infrastructure and market access (Takeshima and Liverpool-Tasie, 2015). These patterns confirm that fertilizer use in SSA is often of limited economic benefit once uncertainty, transaction costs, and a lack of infrastructure investments are accounted for – and that effective interventions require distinguishing farmers by plot size, agroecology, and degree of market integration.

The rationality of low fertilizer use under structural constraints is well-documented. In Ethiopia, fertilizer application rates dropped significantly when farmers faced risks of a poor harvest due to rainfall failure. This was not because fertilizer was unprofitable on average, but because the cost of a failed harvest under liquidity constraints outweighed the expected gains (Dercon and Christiaensen, 2011). Moreover, experiments in Ghana confirmed that when credit and insurance were made available, farmers substantially increased fertilizer use, indicating that structural constraints, rather than limited knowledge, were the determining factor (Karlán et al., 2014). Economic analyses reinforce this picture: in Zambia, fertilizer was profitable at commercial prices for only 37 percent of urea users and as few as 8 percent of those applying basal fertilizer on acidic soils – meaning low use often reflected a correct assessment of



returns (Burke, Jayne and Black, 2017). Similarly in Nigeria, marginal returns to nitrogen on maize decline sharply beyond an application rate of 50 kg/ha meaning that additional fertilizer beyond a modest threshold yielded diminishing economic returns and made those applications unprofitable for many smallholders (Adjognon, Liverpool-Tasie and Reardon, 2017).

Finally, a substantial structural constraint is the price environment itself. Farm-gate fertilizer prices remain high across much of SSA, while food crop prices are significantly lower and often more variable than those in other areas (Maître d'Hôtel and Le Cotty, 2018) – including in Asian countries where the green revolution succeeded (Hernandez et al., 2018; Pingali, 2023). Farmers' fertilizer demand often depends on their ability to sell increased agricultural production at remunerative prices, emphasizing the importance of farmer-consumer market connections. While better infrastructure and logistics can expand market access (Benson and Mogues, 2018), as already discussed, improvements in connectivity alone are insufficient where the underlying price environment renders fertilizer investment unprofitable. The classic characterization of the "poor but efficient" farmer remains apt in many SSA contexts: constrained by poverty, risk, and missing markets, smallholders optimize for survival rather than maximum yield (Dercon and Christiaensen, 2011). Under these conditions, maximizing food security – by preserving cash as a buffer against crop failure – can be a more rational strategy than investing in expensive inputs with uncertain returns.

3.3 SSA RELIES ON FERTILIZER IMPORTS WHILE REGIONAL POTENTIAL UNTAPPED

Sub-Saharan Africa's fertilizer sector is characterized by a marked structural imbalance: heavy dependence on imports to meet domestic demand, limited regional production capacity, and significant — though largely unrealized — potential for greater self-sufficiency. This section examines the region's position within global fertilizer markets, tracing import and export trends, production

disparities between North and sub-Saharan Africa, and the structural barriers and emerging opportunities that will shape the sector's trajectory in the years ahead. The analysis in this section predates the price shocks resulting from the closure of the Strait of Hormuz in 2026, but provides a survey of SSA's structural position in global fertilizer markets.

SSA's reliance on imports for almost all its fertilizer needs creates a direct vulnerability for farmers and food systems in terms of prices and supplies (see Figure 10). Moreover, while SSA depends on overseas suppliers, the North African region is a substantial exporter – particularly Egypt for nitrogen and Morocco for phosphate.

Imports of N, P and K to SSA totalled 4.7 million tonnes in 2022, down from a historic high of 5.3 million tonnes in 2020 (Figure 11). Despite the decrease in import volume, expenditure rose sharply – from USD 3.8 billion in 2019 to USD 5.4 billion in 2022 – as a period of escalating global fertilizer prices meant that SSA countries were paying substantially more for less. This dual burden of rising costs and reduced procurement has direct implications for fertilizer affordability at the farm level and compounds the constraints already discussed (Schmidhuber and Qiao, 2022).

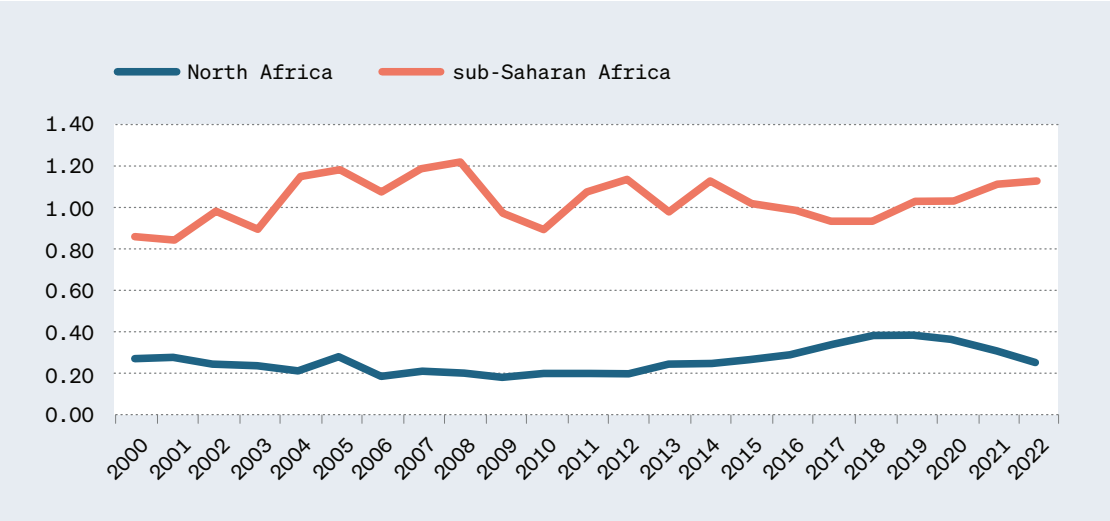
This reliance on imports underscores the low production and export of fertilizers within SSA itself, despite the region's potential. Exports reached a historic high of 1.76 million tonnes in 2022 (Figure 12), with mineral fertilizers from North Africa at six times the volumes exported from SSA. Production of N, P and K in sub-Saharan Africa jumped to a historic high of 1.71 million tonnes in 2022, a figure dwarfed by the 12.2 million tonnes produced in North African countries (Figure 13).

Underlying these disparities is the geographical concentration of mineral fertilizer production in a few countries with the natural resources required for their production – including natural gas, phosphate, and potassium. Just ten countries produce approximately 73 percent of global fertilizer supplies.¹⁵ The leading nitrogen fertilizer producers in Africa are Egypt (46 percent), Morocco (21 percent), Nigeria (17 percent), Algeria (13 percent), South Africa and Tunisia (2 percent each) (see Figure 14). Egypt is seventh among the top global producing countries for nitrogen-based fertilizers – some 3 percent of global production in 2022 (FAO, 2024a). Morocco is the second-largest producer of phosphate fertilizer (P) in the world (12 percent of 2022 production) (FAO, 2024a) and approximately 70 percent of global reserves.

SSA does, however, have significant untapped production potential – though there are substantial structural barriers to realizing it. Nigeria illustrates the opportunity and its limits. It has large natural gas reserves, with three major urea producers with a combined capacity of 5.6 million tonnes, of which half is consumed domestically. It is also a significant producer of NPK fertilizer, but relies for this on key inputs – phosphate from Morocco and potassium from the Russian Federation. Elsewhere on the continent, Togo and Senegal also have appreciable phosphate reserves and Ethiopia and Eritrea share a high-grade but still unexploited potash deposit, while further potash reserves exist in the coastal Congo Basin of the Republic of the Congo.

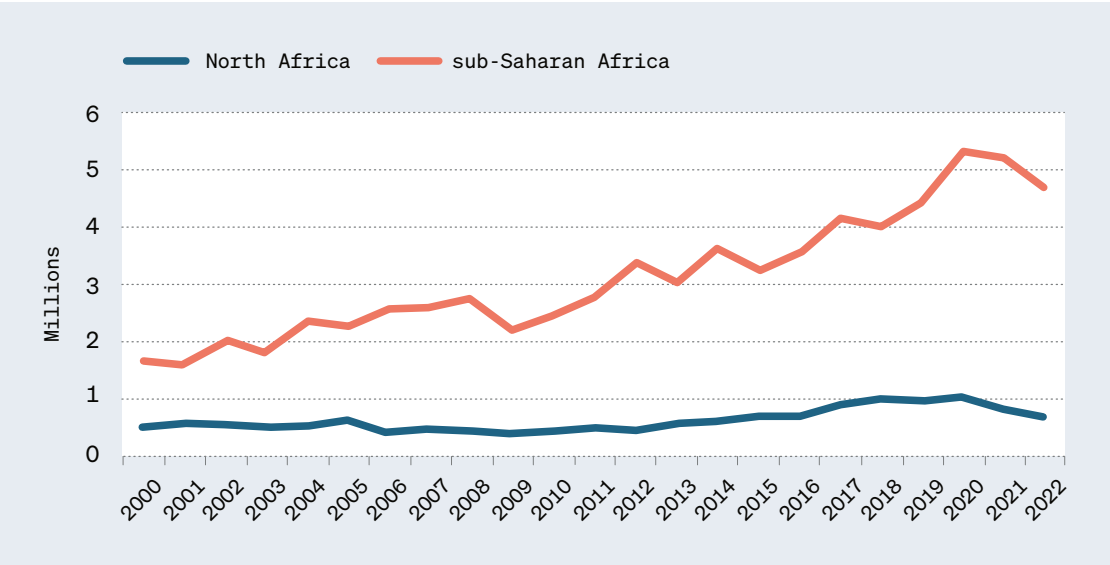
¹⁵ In addition, the top five fertilizer companies represent 57 percent of market share in 2024 (Global Market Insights, n.d.). Moreover, the International Food Policy Research Institute (IFPRI) notes that 57 percent of global trade in phosphates comes from three countries – China, Morocco and the Russian Federation (Hebebrand and Laborde, 2022).

FIGURE 10
 Share of imported N, P and K out of total agricultural use, North Africa and sub-Saharan Africa, 2000–2022



SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN. Licence: CC BY-4.0.

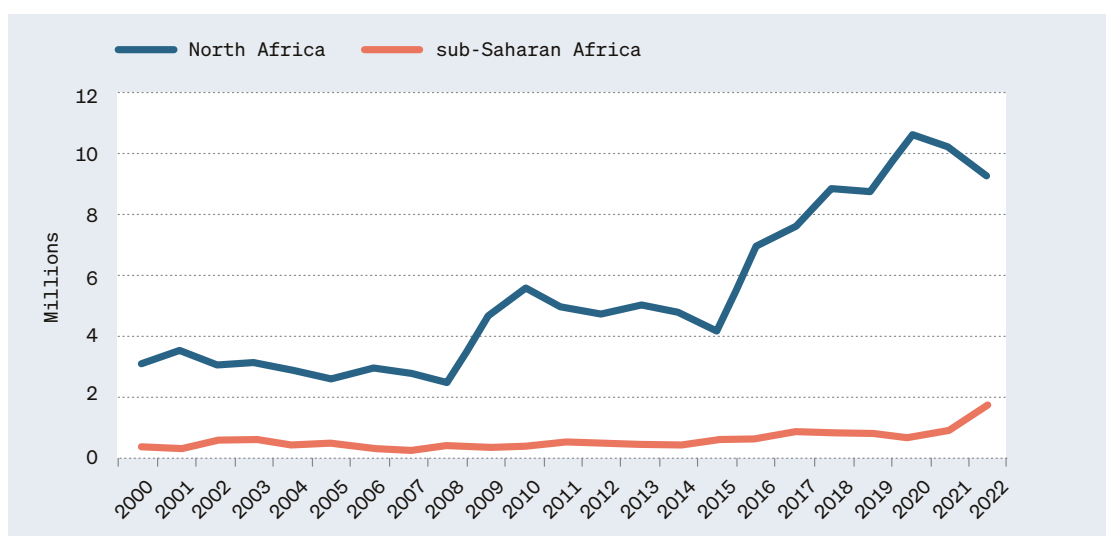
FIGURE 11
 Imports of N, P and K, North Africa and sub-Saharan Africa (tonnes), 2000–2022



SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN. Licence: CC BY-4.0.

FIGURE 12

Exports of N, P and K, North Africa and sub-Saharan Africa (tonnes), 2000–2022

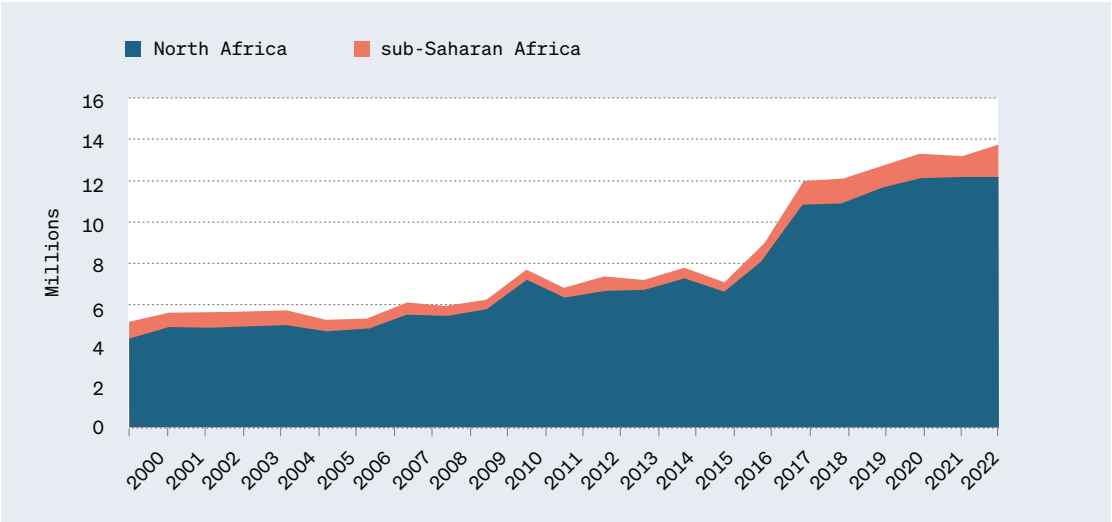


SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN. Licence: CC BY-4.0.

Large-scale production infrastructure is highly capital-intensive and geographically concentrated – exemplified by Morocco's phosphate operations and a major new urea and ammonia plant in Nigeria – and is unlikely to expand across SSA in the near term. A Commercial Agriculture for Smallholders and Agribusiness (CASA) study showed that for such large operations bulk exports are often more commercially attractive than local distribution (CASA, 2023). A more accessible opportunity may lie in processing infrastructure such as blending facilities, which mix granular fertilizers (e.g. N, P and K) to produce customised nutrient combinations. Nearly 100 such plants are active across the African continent (AfricaFertilizer, n.d.), and offer local investors a lower-capital entry point for tailoring fertilizers to specific soil and crop conditions, potentially improving both soil management and yields. Realizing this potential is not straightforward, however, as limited economies of scale and high transport costs may constrain the commercial viability of local blending operations.

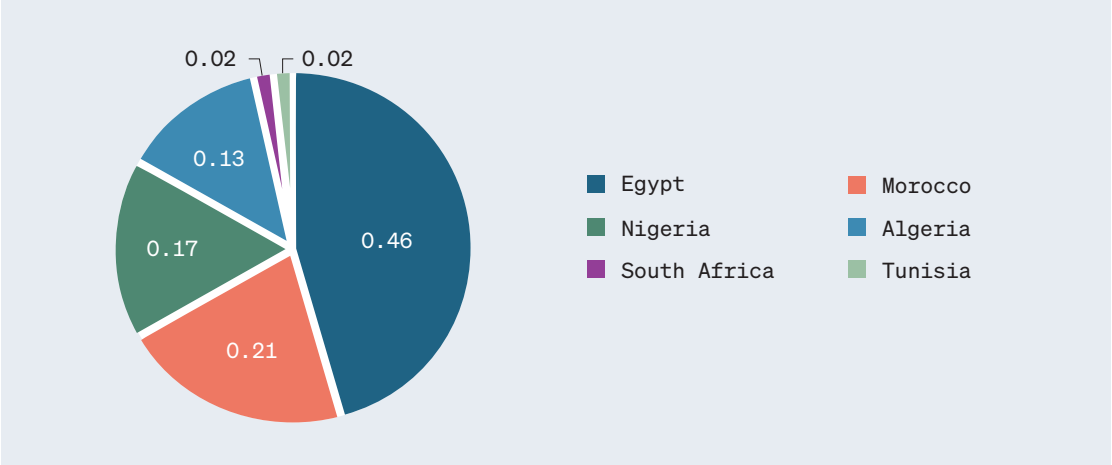
Looking further ahead, green ammonia (NH_3) has attracted attention as a promising pathway for SSA to produce its own nitrogenous fertilizers from local renewable resources, reducing dependence on imports. However, its adoption faces significant technical and economic challenges. Green ammonia uses hydrogen from water electrolysis and nitrogen drawn from the air, powered by 100 percent renewable energy and eliminating the need for natural gas. SSA countries could, in principle, produce their own nitrogenous fertilizers using local renewable resources, potentially transforming the industry while enhancing food security (Vanlauwe et al., 2023). However, green ammonia carries substantially higher production costs than conventional grey ammonia (World Bank, 2025b) – a critical barrier in a region where prices already hold back fertilizer adoption. A further complication is that producing fertilizers such as urea from green ammonia requires an external carbon source, typically supplied through costly direct air capture (DAC) technology, which adds significantly to the final cost of green urea products. For these reasons, green ammonia is unlikely to offer an affordable large-scale solution in the near future (Alimento, 2025).

FIGURE 13
Production of N, P and K, North Africa and sub-Saharan Africa (tonnes), 2000–2022



SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN.
 Licence: CC BY-4.0.

FIGURE 14
Share of total production of N in Africa, 2022



SOURCE: FAO. 2024. FAOSTAT: Fertilizers by nutrient. [Accessed 29 April 2026]. www.fao.org/faostat/en/#data/RFN.
 Licence: CC BY-4.0.



3.4 EMERGING BUT STILL LIMITED MARKETS FOR BIO-BASED FERTILIZERS

Bio-based fertilizers – encompassing both organic fertilizers and biological fertilizers – represent a viable alternative to mineral inputs, and have a particularly important role to play in Africa over the long term, given the continent's high rate of soil organic carbon (SOC) degradation (Maçik, Gryta and Frac, 2020).

Both organic fertilizers, derived from plant and animal residues, and biological fertilizers that incorporate microorganisms fixing atmospheric nitrogen and enhancing nutrient availability, offer broader soil health benefits than just providing nutrients – improving soil structure, supporting microbial activity, and building long-term health. A farmer's decision to use them is nonetheless influenced by socioeconomic factors including land and labour availability, opportunity costs, the degree of crop-livestock integration, and the cost of such bio-based inputs (Vanlauwe et al., 2023).

The African organic fertilizer market is projected to grow from an estimated USD 560 million in 2025 to USD 990 million by 2029 (Mordor Intelligence, 2024). This growth is fuelled in part by substantial expansion in organic farming, the benefits of organic fertilizers in conventional agriculture beyond the provision of N, P and K, and rising mineral fertilizer prices that have prompted some farmers to shift towards organic alternatives. Organic agriculture in Africa has grown from around 50 000 hectares in 2000 to approximately 2.03 million hectares in 2019, though the continent still has the smallest share of organically farmed land of any world region (Statista, 2024). Africa's organic fertilizer market is segmented by type (i.e. microorganisms and organic residues), application, and geography, with key markets comprising South Africa, Egypt and Ethiopia (Mordor Intelligence, 2024).

The global market for biological fertilizer is expanding in several advanced economies, supported by tax incentives, input subsidies, as well as market development and financial support. SSA countries will need to make a similar strategic shift towards biological fertilizer in order to realize the economic and environmental potential of bio-inputs observed in regions such as Latin America (Bullor *et al.*, 2024). Bio-inputs are a central element in the transition to sustainable agrifood systems enabling the conservation of biodiversity, increasing soil health, and enhancing agricultural efficiency. The bio-inputs market in Latin America, growing at approximately 15 percent annually, reflects their role in reducing farmers' reliance on costly conventional agrochemicals and in maintaining access to highly regulated export markets. However, realizing this potential requires overcoming constraints including the need for specialized cold storage and distribution infrastructure, and the establishment of robust national regulatory frameworks to ensure product quality and build farmer trust (Bullor *et al.*, 2024).

In SSA, the potential of biological fertilizer products has been largely untapped due to inadequate market development, weak regulatory frameworks, poorly enforced quality control standards, and limited farmer knowledge and skills in using these inputs. These challenges have contributed to unregulated production and an influx of commercial biofertilizers of questionable quality – undermining farmer confidence and hampering adoption across the continent (Raimi, Roopnarain and Adeleke, 2021). Addressing these barriers requires policy commitment to train extension agents, strengthen standards and enforcement, and mobilize private sector investment and compliance with established regulations.

Such steps could help to address the high rate of degradation of SOC – the crucial carbon component in soils (Dugué, Andrieu and Bakker, 2024). Living inocula of microorganisms that fix atmospheric nitrogen and enhance nutrient availability can be reused from one season to the next – an inexpensive practice widely used in India and Latin America that requires less technology and knowledge than high-precision fertilization (Umesha, Singh and Singh, 2018). Such efforts should, however, be accompanied by quality and safety assessments to ensure that bio-based products meet agronomic standards and do not introduce contaminants or pathogens into soils and food systems.

SOIL, AGRONOMIC, AND ECONOMIC CONSTRAINTS ON FERTILIZER EFFICIENCY

The efficiency of mineral fertilizer use in SSA is shaped by a variety of interlocking agronomic, environmental and economic factors. Fundamental to these are soil characteristics, agricultural practices, and the fertilizer type, quality, amount and timing of application.

In general, deep soils rich in organic matter content with good water retention offer greater productivity and make fertilizers more effective, as more nutrients are available for optimal crop growth. Africa's more common sandy or ferritic soils, by contrast, are frequently acidic – whether naturally or as a result of poor soil management – limiting macro- and micronutrient uptake by crops. Soil acidity affects approximately 25 percent of soils in Africa (FAO and ITPS, 2015b),¹⁶ representing one of the main biophysical constraints on fertilizer response.

Farming practices significantly compound these soil constraints, with nutrient mining a widespread concern across the continent. This process – characterised by an imbalance between soil nutrient inputs and outputs – depletes nutrient reserves and SOC, contributing to erosion, acidification, and the release of CO₂ and nitrous oxide into the atmosphere. In SSA, this process is driven primarily by the chronic underuse of mineral fertilizers, leaving essential nutrients unreplenished (Vanlauwe *et al.*, 2023). This stands in contrast to conditions in other regions, where overuse of mineral fertilizers drives acidification, soil pollution,¹⁷ water quality hazards including eutrophication, and risks to human health such as elevated nitrates in drinking water.

Improving nitrogen use efficiency will become increasingly important in SSA's agrifood systems. Globally, nitrogen use is inefficient in that plants do not make adequate use of the applied nitrogen, implying that the increase in yield per unit of nutrient applied is not as high as it could be. Less than half of the nitrogen supplied to crops is utilized on average, and research suggests that nitrogen applied to wheat, rice, and maize could be reduced by 14–29 percent worldwide while maintaining yields (West *et al.*, 2014). Greater efficiency is achievable through changes in fertilizer timing, placement, rate and type (West *et al.*, 2014). For African producers, improvements in nutrient use efficiency would directly reduce input costs – a significant consideration given the price constraints explored earlier (Vanlauwe *et al.*, 2023).

The efficiency of fertilizer use in SSA is further complicated by diverse cropping systems and by the geographical and economic conditions facing producers. Maize, for example, is agronomically highly responsive to nitrogen fertilizer, and its cultivation is associated in many places with greater mineral fertilizer use. From an economic point of view, however, whether that use is profitable depends on a range of factors beyond agronomy and affecting producer prices – including road infrastructure and transport costs, fertilizer and other input prices, output market prices, and market structure.

¹⁶ Acidic soils are deficient in phosphorus, calcium and magnesium, and may have toxic levels of aluminium.

¹⁷ Between 30 percent and 50 percent of mineral fertilizer nutrients end up contaminating groundwater or volatilizing into the air (Chojnacka, Moustakas and Witek-Krowiak, 2020).





Growing opportunities to improve soil health in SSA

4.1 PRODUCTION POTENTIAL IN HEALTHY SOILS

Approximately 30 percent of arable land in Africa has good agronomic potential, suitable for intensification through mineral or bio-based fertilizers (FAO and ITPS, 2015a). In many humid, sub-humid, and some semi-arid areas, a “doubly green” revolution is technically feasible – improving both the environment and agricultural output simultaneously.¹⁸ Assuming there is water and organic matter in the soil, using more fertilizer in an efficient manner would help to increase yields and output.

Mineral fertilizers are needed to reach yield potential. In case studies across a range of SSA countries, Falconnier *et al.* (2023) found that maize yield potential ranged from 3.5–11.9 tonnes/ha (with a median of 6.5 tonnes/ha), implying median N uptake of 113 kg/ha. They argue that N, P and K needs at these yield levels in smallholder systems cannot be met through biological nitrogen fixation, such as by legumes, and animal manure alone (Falconnier *et al.*, 2023). Nevertheless, from a nutrient balance perspective, mineral fertilizer remains necessary in these cases to improve soil fertility and increase yields in the short term (Falconnier *et al.*, 2023; Vanlauwe *et al.*, 2023), even when applying agroecological principles to increase nutrient availability.

4.2 THE POLITICAL ECONOMY AND LIMITATIONS OF FERTILIZER SUBSIDIES

Low fertilizer use rates and the need to boost agricultural production have led to calls over the last two decades for African governments to take more action. Three successive regional declarations – the Comprehensive Africa Agriculture Development Programme’s (CAADP) Maputo Declaration of 2003, the 2006 Abuja Declaration, and the 2014 Malabo Declaration – each committed member states to increasing fertilizer use and agricultural productivity, yet there has been limited impact.

¹⁸ This differs from the green revolution, which historically focused on agricultural output (Conway, 1997).

CAADP committed African governments to double agricultural productivity by focusing on the provision of improved inputs. The Abuja Declaration declared fertilizer a strategic commodity and resolved that African Union member states should promote its use through targeted subsidies (AUC, 2006). That declaration was most specific, targeting fertilizer consumption of 50 kg/ha by 2015, against an SSA average of just 8 kg/ha in 2006 (AUC, 2006). It set out interventions at regional and country levels – including policy and regulatory frameworks, targeted subsidies, and improved access to complementary inputs – and was reinforced in 2014 by the Malabo Declaration in which African leaders affirmed their CAADP commitments and called for smart subsidies carefully targeted to smallholder agriculture (AUC, 2014).

Some countries responded by revising their fertilizer policies and regulations (among them Burkina Faso, Ghana, Kenya, Mali, Mozambique, Rwanda, Uganda, the United Republic of Tanzania, and Zambia) (Abate, Abay and Spielman, 2020), but many still lack a coherent fertilizer policy (AGRA, 2019). For most, subsidies have been the preference in policy and programmes.

Fertilizer and input subsidies in Africa are estimated at 14–26 percent of national budget allocations to agriculture between 2011 and 2014 (Jayne *et al.*, 2018), a substantial share of agricultural spending in some of SSA's biggest economies. Despite varying programme designs across countries, crops and years (see Box 2), their overall impact on productivity and poverty outcomes has been disappointing.

Most studies in SSA show that fertilizer subsidies do not significantly impact production, land productivity or farmers' incomes. They are generally costly and poorly targeted – particularly in landlocked countries (Arslan *et al.*, 2022; Maître d'Hôtel and Issoufou, 2018) – and often subject to rent-seeking (Holden and Lunduka, 2012). Malawi's subsidy experience is regularly cited, though it remains contested in terms of both implementation and outcomes (Denning *et al.*, 2009; Dorward and Chirwa, 2011; Sachs, 2012; Jayne and Rashid, 2013). More fundamentally, the heterogeneity of smallholder farming systems and soil conditions demands targeted rather than blanket recommendations – a reality that broad subsidy programmes have struggled to accommodate (Vanlauwe *et al.*, 2023).¹⁹

The political economy of these policies and programmes also adds another layer of complexity. In SSA, fertilizer subsidies are widely believed to be popular among voters, making them attractive to politicians seeking to secure or retain power – even though evidence for this effect is mixed (Mason, Jayne and Van de Walle, 2017). Even the best-designed fertilizer policies and subsidy programmes are difficult to implement and can produce unintended consequences.

The lesson from this experience is that SSA governments cannot achieve productivity goals by simply increasing volumes of fertilizer distributed or public funds allocated (Abate, Abay and Spielman, 2020). More effective, holistic policies and programmes are needed that improve agricultural productivity while carefully considering socioeconomic, agronomic, and sustainability

¹⁹ Fertilizer subsidies in Asian countries have proven much more effective because of a more effective food market in particular. Between 2010 and 2022, the Government of India provided an average subsidy of USD 228.5, USD 110.4, and USD 70.5 per tonne for urea, DAP, and MOP, respectively. As a result, farmers paid only 22 percent of the international market price for urea, and 71 percent and 73 percent for DAP and MOP, respectively (TAAS, 2023).

Public support for fertilizer use in SSA has taken three broad forms, broadly in line with the evolution of agricultural policies (Lehe *et al.*, 2025).

The first entails state-led fertilizer marketing, characterized by universal or near-universal fertilizer price subsidies, with public agencies prominent in procurement, pricing, and distribution. Prevalent across much of Africa from independence in the 1960s to the structural adjustment programme era in the 1980s, these policies increased fertilizer use among farmers – particularly in the later years of this period (Eicher, 1995; Byerlee and Eicher, 1997). The design and implementation of these policies have, however, been shown to have high fiscal cost, with market-distorting effects and significant rent-seeking (Jayne and Jones, 1997; Morris *et al.*, 2007; Yamano and Arai, 2010).

The second is market-oriented reform, in which governments introduced more market-led fertilizer procurement and distribution mechanisms. These reforms have varied considerably in scope and depth across countries, and their results were mixed – and often controversial (Jayne *et al.*, 2002, 2003; Minot and Benson, 2009).

The third category covers policy and programme modification, aimed at improving targeting and mitigating the unintended consequences – budgetary, environmental, and market – of earlier approaches. In the late 2010s, Nigeria introduced an innovative mobile phone-based input subsidy programme delivering fertilizer subsidies through electronic vouchers. Other countries, including Malawi, Mozambique, Rwanda, Zambia, and Zimbabwe, have been following this lead with e-wallet and e-voucher schemes. The effectiveness of these innovations and smart subsidy programmes will depend on the institutional and political contexts in which they are introduced (Jayne *et al.*, 2018).

Despite this evolution, the evidence increasingly points to the limits of fertilizer subsidies as a standalone tool to sustainably improve productivity. Subsidies consistently fail to address the underlying constraints that drive under-use: poor soil health reduces returns to standard fertilizer packages; rainfed systems make fertilizer investments weather-dependent and risky; inefficient output markets keep crop prices too low relative to input costs; and underdeveloped financial markets prevent farmers from purchasing complementary inputs or insuring against production risk (Hill and Resnick, 2025). Cross-country incidence analysis also shows that subsidy benefits accrue disproportionately to better-off farmers, while the programmes often crowd out private input market development.

These findings point toward a reorientation of public spending, rather than incremental reform. Portions of subsidy budgets redirected toward roads, irrigation, extension services, ISFM, and private sector market development are likely to address the underlying constraints more effectively – and to make any remaining subsidy spending more productive in the process (Hill and Resnick, 2025). This logic underpins policy developments since the Abuja Declaration and is further articulated in the 2024 Nairobi Declaration on Fertilizer and Soil Health, which called for a more integrated approach to agricultural input policy.

SOURCE: Authors' elaboration based on Abate, G.T., Abay, K.A. & Spielman, D.J. 2020. Fertilizer policies and implications for African agriculture. In: D. Resnick, X. Diao & G. Tadesse, eds. 2020 Annual trends and outlook report: sustaining Africa's agrifood system transformation. Chapter 4, pp. 33–44. Washington, DC and Kigali, International Food Policy Research Institute (IFPRI) and AKADEMIYA2063. https://doi.org/10.2499/9780896293946_04

The challenge is less one of insufficient demand, but rather the high transaction costs and infrastructure gaps that isolate many smallholder farmers from these growing markets.

trade-offs between land productivity, labour productivity and inner-household labour allocation, soil health and fertility and overall smallholders' livelihoods. Evidence points to the need for governments to address the challenges of underinvestment in agrifood systems, including crop breeding, agronomy, irrigation, soil testing, and extension services, as well as the market and infrastructure connections that together determine whether fertilizer use is profitable for smallholders. Public agricultural R&D spending in SSA was estimated to stand at just 0.39 percent as a share of agricultural GDP in 2016, far below the African Union and United Nations recommended minimum of 1 percent (ASTI, 2020). Moreover, questions remain about how effectively even that limited funding has been prioritized, managed and used. Current policies and incentives may also not adequately encourage private sector investment in agricultural innovation (Fuglie *et al.*, 2019).

The green revolution in Asia offers important, context-specific lessons for SSA. Writing about the relevance of Asia's experience for agricultural growth today, Pingali (2023) notes that political economy imperatives – particularly the management of urban food supplies and food prices – played a central role in driving productivity gains, alongside output price supports, input and credit subsidies, and controls on the international food trade. These combined to incentivize farmers to invest in high-yielding technologies and practices.

SSA, however, presents a different landscape today, with large increases in urban food consumption and the rapid expansion of dynamic agrifood value chains across Africa (Reardon, 2015). The challenge is less one of insufficient demand, but rather the high transaction costs and infrastructure gaps that isolate many smallholder farmers from these growing markets. In remote areas with limited market access, production increases can lead to local surpluses and price drops, undermining farmers' incentives to invest in fertilizers and improved practices (Le Cotty, Maître d'Hôtel and Ndiaye, 2017).

These differences suggest that smart targeting of fertilizer subsidies should prioritize farmers with better access to dynamic urban markets or integration into robust value chains, whether for domestic consumption or export. This approach builds on successful precedents such as cotton value chains that drove agricultural intensification in parts of West Africa decades ago. Building on this will also require investment in rural infrastructure, market information systems, and value chain development – providing the conditions where fertilizer use becomes economically viable for a wider range of smallholders (Le Cotty, Maître d'Hôtel and Ndiaye, 2017).



4.3 FROM ABUJA TO NAIROBI: CONSENSUS ON INTEGRATED SOIL HEALTH

Over the past two decades, Africa's political commitment to fertilizers, and soil fertility more broadly, has undergone a significant shift. Starting from a narrower focus on boosting mineral fertilizer consumption in the 2006 Abuja Declaration, the highest-level continental commitments now take in a broader, more integrated vision of soil health and sustainable agrifood systems, culminating in the 2024 Nairobi Declaration.

The Africa Fertilizer Summit held in Abuja, Nigeria in 2006 identified the critical need to increase fertilizer use to stimulate agricultural productivity growth and reverse the trend of soil nutrient depletion. Rooted in the productivity vision of the East Asian Miracle three decades earlier (World Bank, 1993), the Abuja Declaration treated mineral fertilizer as the primary lever for intensification and in delivering an “African Green Revolution”. There was little mention of soil health, nor of the possible negative health and environmental impacts of the green revolution in Asia (Pingali, 2012). As explored in Section 4.2, the results were limited, and even with the reinforcement of the Malabo Declaration in 2014, fertilizer use in SSA remained well below target. Moreover, the structural constraints on smallholder investment in inputs largely went unaddressed.

The food and climate crises that have deepened since 2006 have had a remarkable impact on the policymaking context in Africa – and the thinking on sustainable agrifood systems. Eighteen years after Abuja, the 2024 Africa Fertilizer and Soil Health Summit (AFSH), convened in Nairobi, Kenya, explicitly reframed the agenda, as an opportunity for “decisive steps towards sustainably increasing agricultural productivity to reduce hunger and malnutrition, improve livelihoods, and adapt to the impacts of climate change” (African Union, 2024a; AUDA-NEPAD, 2024; African Union, 2024b). These words marked a significant political evolution: Not only was soil health now explicitly embraced in the 2024 summit title, but climate change – entirely absent from the 2006 declaration – was now central to the agenda. While the 2006 declaration had briefly acknowledged the environment, there had been no mention of sustainability or sustainable approaches.

The shift was clear in the Nairobi Declaration – which balances the productivity narrative with convincing evidence on the importance of soil health, soil nutrients and soil fertility – and the Africa Fertilizer and Soil Health Action Plan: 2024–2034.²⁰ Crucially, the action plan acknowledges that “mineral fertilizers alone are inadequate to reverse land degradation and sustainably increase the productivity of Africa’s soils” (African Union, 2024d). ISFM and sustainable soil and water management are set out as the framework for future investment.²¹ Leaders recognized the integrated nature of Africa’s agricultural sub-sectors and soil health concerns, and the need to prioritize integrated soil and water conservation and management at watershed, landscape, and catchment level. In the spirit of the Africa Common Position on Food Systems (African Union, 2023), the declaration articulates a comprehensive aspiration to improve food security and nutrition, promote inclusive economic growth, and ensure equitable and balanced territorial development while ensuring environmental sustainability and resilience. The declaration also broadened the goal from crop productivity and profitability to encompass socioeconomic and environmental sustainability, considering climate change adaptation and mitigation, rehabilitation of degraded land, and restoration and maintenance of ecosystem services, including biodiversity. The Nairobi Declaration reiterates that increasing fertilizer use in Africa remains a valid goal to be achieved in a way that supports soil health and environmental resilience. The measure of progress will lie in how effectively these commitments are translated into coordinated actions.

The 2025 Kampala CAADP Declaration moved swiftly to embed this agenda within the continent’s principal framework for translating agricultural undertakings into national investment plans, reaffirming core commitments to investing in soil fertility and health as the cornerstone of sustainable agrifood system transformation. African Union member states pledged to scale up investments in ISFM and promote climate-smart and regenerative practices alongside the enhanced use of bio-based and mineral fertilizers in an environmentally sustainable manner. Critical to this effort is recognition of the necessity of improving farmer access to quality, affordable fertilizers through strengthened functional input systems, regional harmonization of standards, and boosting local production capacity, while mitigating post-harvest losses and soil

²⁰ Endorsed by the Heads of State and Government of the African Union during the 2024 Africa Fertilizer and Soil Health Summit in Nairobi.

²¹ ISFM addresses the concept of production systems in which efficient fertilizers, both mineral and organic; other inputs, such as improved seeds; water use efficiency for irrigation; and other aspects of soil health and sustainable soil and water management are crucial to food security and agricultural sustainability.



degradation. The CAADP Action Plan (2026–2035) focuses on strengthening fertilizer supply chains and distribution, expanding local production, and promoting best application practices through targeted training and subsidies, while pairing fertilizer use with essential soil health monitoring and extension services to ensure balanced nutrient application and long-term productivity (African Union, 2024c).

Alignment between the AFSH and CAADP action plans will be critical to putting the Nairobi Declaration into practice and ensuring action in SSA countries. How that can be achieved, and that investment and policy instruments can help to drive this, is the focus of Section 5.



Policy and investment options

Improving soil health and sustainably boosting food production in SSA require more than better access to fertilizer. They demand a coordinated policy response across four interconnected objectives: improving soil health and fertility; increasing fertilizer availability; improving fertilizer access and affordability; and strengthening producers' incentives to invest in their land. These mutually reinforcing priorities reflect the analysis in this report and should be considered together as part of an integrated soil health and fertilizer policy framework. They are grounded in the 2024 Nairobi Declaration and support the Africa Fertilizer and Soil Health Action Plan: 2024–2034 in the aim to drive the sustainable transformation of agrifood systems across the region.

Secure land tenure is a critical prerequisite for implementing most agricultural investment projects in SSA (Cotula and Toulmin, 2008), as it is fundamental to attracting finance. Land serves as a key asset that can be used as security for loans, enabling farmers to access credit for the expansion of production, the purchase of equipment, fertilizers, and soil improvement inputs. Without secure land rights, farmers face significant barriers to obtaining capital from banks and other lenders, limiting their ability to invest in productivity-enhancing measures and ultimately undermining food security and livelihoods. In short, land tenure determines not only where to invest but also whether to invest, as financial institutions and insurers are unlikely to provide credit or coverage without clear and enforceable land rights.

5.1 IMPROVING SOIL HEALTH AND SOIL FERTILITY

Strategies to prioritize soil health and fertility must be carefully tailored to context. Key considerations include climate variations – distinguishing between dry and humid conditions – soil condition, the interactions between crop and live-stock systems, and the potential for integrating agroforestry. Effective strategies demand nuanced understanding of local soil needs and strategic use of targeted tools and instruments to achieve comprehensive soil health objectives.

There is broad consensus on the need for sustainability in soil fertility, drawing efficiently on both bio-based and mineral sources of nutrients. The Africa Fertilizer and Soil Health Action Plan: 2024–2034 proposes ISFM, while agroecological approaches have been recommended by a range of global institutions (HLPE, 2019; CFS, 2021b). Diverse actors across SSA emphasize different aspects of this broad agenda. Some advocate a transition without chemical inputs, focusing on the bio-based sector; others argue for a balanced approach, particularly in areas where mineral fertilizer use remains low, and where these inputs could complement nutrients provided by agroecological practices such as the use of legumes, manure, biofertilizers, biostimulants, and cover crops. On degraded soils, where mineral fertilizer use is least efficient, restoring soil organic matter and improving water retention are prerequisites, as discussed in Section 2.2. These approaches take time to generate returns – often representing a major constraint for small-scale producers – which underlines the need for policies and incentives that support investment in soil health, not just access to fertilizers (Vanlauwe *et al.*, 2023).

Support for localized soil testing facilities, along with the promotion and supply of related products such as lime for acidic soils and crop- and soil-specific fertilizer blends, is an important aspect of delivering localized solutions at scale. Improving producers' awareness of alternative tools – including portable soil and plant sensors and soil management applications – and of the benefits of different production practices, is a relevant complement in many countries and systems. These efforts work best when combined with incentives for agribusinesses to promote and distribute supplementary products, ideally as part of a bundled package that includes soil and plant analysis alongside other technology-based services.

Recent success stories in localized soil management taken to scale include the AKILIMO site-specific fertilizer recommendation portal for cassava, developed by the African Cassava Agronomy Initiative, and the SOILS Consortium Space to Place programme in Ethiopia, funded by USAID (Vanlauwe *et al.*, 2023). The FAO Soil Mapping for Resilient Agrifood Systems Programme (SoilFER), implemented in Ghana, Kenya, Mozambique, Tunisia, and Zambia, illustrates the potential of an integrated institutional capacity development approach. It combines soil surveys, national soil information systems, strengthened soil laboratories, and farmer-focused extension services, through the Global Soil Doctors Programme. These models demonstrate that strategic investment in localized soil infrastructure can meaningfully improve decision-making on fertilizer use and sustainable soil management (FAO, 2024c). High-quality soil health advisory services that deliver practical, accessible information help to build producer trust and improve uptake.

Strengthening soil science and plant nutrition knowledge across the full range of stakeholders – from farmers and extension agents to researchers and agribusiness professionals – is an important foundation for sustainable

agriculture. This requires expanding and upgrading soil science education through formal institutions and lifelong learning, including informal education in local languages (AUC, 2023). Proven models suitable for scaling up include the Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA) and the Centre of Excellence for Soil and Fertilizer Research in Africa (CESFRA) at Mohammed VI Polytechnic University in Morocco.

Soil fertility investments may be more effective when differentiated or targeted by farm type. Subsistence farmers with limited land and poor market access face fundamentally different constraints to those of semi-commercial or commercial producers, and uniform approaches will consistently underperform relative to targeted ones. For smallholder subsistence farmers cultivating less than one hectare primarily for family consumption, programmes that reduce risk and build economic resilience – such as drought insurance, cash transfers, productive safety nets, and access to credit – should precede or accompany fertilizer promotion, rather than implicitly assuming that the necessary enabling conditions for fertilizer use are already in place. For emerging commercial farmers with larger plots and better market connectivity, the priority shifts toward improved market access, price stabilization, and tailored fertilizer blends or credit programmes that make intensification economically viable. Sequencing and targeting interventions in this way is not just more equitable – it is more likely to generate sustained productivity gains.

5.2 INCREASE THE PRODUCTION AND AVAILABILITY OF FERTILIZERS FROM SSA

The second policy objective concerns increasing the availability of both mineral and bio-based fertilizers in SSA. This could be pursued through four complementary approaches: (i) boosting local production for domestic and regional supply; (ii) improving incentives and reducing risks for private sector investments (e.g. green hydrogen-based fertilizers²² and bio-based production); (iii) addressing structural market inefficiencies and strengthening competition to achieve economies of scale, including through improved transport infrastructure and reduced trade-related administrative burdens; and (iv) prioritizing foundational public investments to improve overall sector performance.

Promoting investment in production and adoption of bio-based fertilizers in particular could have a significant impact on soil health and crop productivity. Where farmer demand for fertilizers remains unmet, a fertilizer production policy would seek to remove barriers to national and regional trade and offer incentives to increase the supply and improve the price competitiveness of African-produced fertilizers. Assuming there is unmet demand from farmers, a comprehensive policy could also aim to ease market constraints hampering producer profitability and to incentivize the production and distribution of bio-based and mineral fertilizers.

Rural fertilizer markets in SSA do not function efficiently, suggesting that strategic interventions are necessary. These include measures to promote market competition, mitigate risks for private investors, improve distribution systems, and strengthen private sector participation in the fertilizer value chain. Investment in enabling public goods – such as roads, storage facilities, information systems, and extension services – is often a critical prerequisite to

²² Promoting decentralized ammonia fertilizer production can play a crucial role in increasing fertilizer availability, thereby enhancing food security. This cross-cutting technology is rapidly evolving and becoming increasingly cost-competitive.

well-functioning markets, creating the favourable economic environment in which both locally produced and imported fertilizers can reach farmers efficiently.

Strategic market mechanisms and financial instruments can address supply chain inefficiencies and price volatility directly. The African Trade Exchange Platform, a business-to-business and business-to-government e-commerce platform, was launched in 2022, exemplifies this approach by aggregating continental demand to negotiate competitive prices and enable access to affordable fertilizers, creating economies of scale and reducing transaction costs. Similarly, the Africa Fertilizer Financing Mechanism (AFFM), a fund established in 2007 and administered by the African Development Bank Group (AfDB), provides critical financial support to enhance pan-African agricultural productivity by promoting fertilizer use. Combining AFFM credit guarantees with the African Trade Exchange Platform has the potential to mitigate some of the market failures observed in Africa's fertilizer market.

Realizing the full potential of these mechanisms, however requires substantial public investment in research and development and in innovative financing instruments – such as long-maturity lines of credit – channelled through microfinance institutions and first-tier commercial banks with the explicit aim of improving soil fertility and fertilizer use efficiency among small-holders. Concerted efforts to engage more actively with alternative fertilizer businesses, such as early-stage ventures in biological products or larger-scale projects in green ammonia production, conventional manufacturing and blending facilities, are equally necessary to build the diversified supply base that agrifood systems need.

5.3 INCREASE ACCESS TO AND AFFORDABILITY OF FERTILIZERS

A range of options could be used to improve farmers' access to fertilizers and address the constraints identified. Indirect approaches include investments in infrastructure and a competitive transport market that reduces delivery costs, resulting in lower farm-gate fertilizer prices.

Governments may consider smart fertilizer subsidies combined with other incentives as a policy option for increasing farmers' access to fertilizers and other soil nutrients. However, the impact of such subsidies is highly conditional on soil health, market access, weather reliability, and credit availability. Many of these conditions may be absent for the most resource-constrained farmers, and previous programmes have fallen short because they were not in place (Hill and Resnick, 2025). Redesigning smart subsidy models can improve the efficiency of fertilizer use and restore soil fertility – provided they combine support for bio-based and mineral fertilizers with incentives to invest in complementary practices. Indeed, they may prove more sustainable than price support programmes alone. However, in remote areas where infrastructure and market access are major constraints, subsidizing fertilizer may be the least efficient use of public funds. Where conditions are more favourable, targeted subsidies could serve as investments facilitating access to complementary products and practices that enhance nutrient use efficiency and restore soil fertility (Vanlauwe *et al.*, 2023).

A further consideration is whether a portion of current subsidy spending could be repurposed toward investments that generate higher and more durable development benefits. For repurposing to be politically viable, alternative investments should share the features that make subsidies attractive – immediacy of benefits, visibility, and the ability to target specific beneficiaries.



Sequencing matters: the withdrawal of existing subsidies and the scaling of alternatives must be carefully coordinated, with continuous monitoring of impacts used to refine targeting, assess soil and climatic conditions, and ensure budget efficiency (Hill and Resnick, 2025).

Improving producer access to credit can also facilitate access to fertilizer, as shown in the West African cotton systems characterized by interlinked credit, input and output markets (Tsakok and Guedegbe, 2019; Tefft, 2010). Another policy instrument is developing inventory credit, which enhances production and fertilizer use by alleviating liquidity constraints at the onset of the planting season, the critical period for access to inputs.

5.4 INCREASE PRODUCERS' PRODUCTION INCENTIVES

The fourth objective consists of improving the incentives needed for producers to invest in fertilizer and other productivity- and sustainability-enhancing measures. A broad empirical literature confirms that smallholder farmers in SSA face multiple, reinforcing constraints – insecure land tenure, lack of access to credit, lack of appropriate research and extension, insufficient transportation and market infrastructure, limited ability to manage risk, increased vulnerability to climate change, weak collective action before markets and public power/institutions, and a lack of access to complementary forms of income (Lowder, Bhalla and Davis, 2025; Stewart *et al.*, 2017). Critically, these constraints tend to reinforce one another rather than operate independently, so that relaxing a single constraint in isolation may often yield limited results (Vanlauwe *et al.*, 2010).

For example, fertilizer subsidies introduced without improvements in output markets tend to have modest and short-lived effects. Extension efforts unaccompanied by access to affordable inputs may change awareness without changing behaviour. Soil restoration practices requiring upfront labour or delayed returns are rarely adopted without income buffers or risk-mitigation mechanisms. Effective production incentive policies therefore need to be designed as coordinated packages, not sequential single-constraint fixes (Hill and Resnick, 2025).

Remunerative farm-gate prices represent a cornerstone of an improved incentive environment. Long-term price support programmes, if maintained and consistently funded, can provide producers with the investment signals needed to intensify production sustainably. However, price support programmes often have a less-than-ideal track record and may pose an institutional risk of budgetary unsustainability due to substantial financial outlays, particularly for public procurement and storage. Historical experience has also shown that governments often lack the necessary resources to sustain minimum prices, especially in the face of significant downward pressure on those prices.

Governments may also need to enhance the provision of essential public goods. Priority investments include rural infrastructure – particularly roads and storage facilities – that reduce transportation costs and connect farmers to markets. Harmonization of regional and sub-regional fertilizer regulations, including quality assurance processes, norms and standards, and cross-border trade facilitation, represent additional public measures to make fertilizer more accessible and more economically viable for African farmers.

Crop insurance represents another option for de-risking farmers' production environment and encouraging investment and use of productive inputs. To date, liquidity constraints before planting, as well as diverse technical factors, have restricted the adoption of crop insurance in sub-Saharan Africa. Subsidizing insurance premiums could make it more accessible to all farmers, or to focus on the most vulnerable, significantly increasing uptake.

Beyond price support and producer insurance programmes already mentioned, a set of complementary investments has shown particular promise in generating immediate, tangible, scalable benefits. These include digital advisory services, which provide timely, personalized guidance to producers, and have been demonstrated to increase farmers' profits in Nigeria. Additionally, payments for ecosystem services can incentivize farmers to adopt environmentally friendly practices, such as shown in Ethiopia. Likewise, support to small-scale irrigation investments that reduce risks and enhance resilience may be equally advantageous investments (Hill and Resnick, 2025). Each of these may be effective in its own right, and may also address the underlying constraints that limit fertilizer response, meaning they can make broader soil health investments more productive, rather than substituting for them (Hill and Resnick, 2025).

5.5 CROSSCUTTING ISSUES FOR POLICY ACTION

In addition to the four policy objectives outlined in the previous sub-section, five cross-cutting issues demand attention: (i) an enabling environment; (ii) innovative extension and farmer knowledge systems; (iii) investment in research and development; (iv) enhanced de-risking instruments for farmers from financial institutions; and (v) robust monitoring and evaluation systems, accountability mechanisms and long-term, multi-sector planning frameworks.



Also deserving of attention is the need to address the issue of land tenure security. First, establishing an enabling environment and regulatory framework is crucial for advancing sustainable soil health, fertility and fertilizer use in SSA. This requires strategic investment across all policy instruments to create conditions in which the private sector can thrive. Key tools include public-private partnerships that align private sector goals with national soil health priorities, supported by incentive-based policies that encourage both public and private investment. Regulatory frameworks should be designed to promote both the availability and responsible use of bio-based and mineral fertilizers across the continent. Engaging farmers and private sector stakeholders in policy dialogue and design can accelerate the adoption of sustainable practices such as agroecological intensification.²³ This approach, when supported by effective policy tools, can deliver economic, social, and environmental benefits through improved nutrient management, crop productivity and nutrient use efficiency in SSA (Vanlauwe et al., 2023).

²³ Agroecological intensification prioritizes on-farm biodiversity conservation and natural regeneration while reducing emphasis on soil tillage and chemical inputs. This approach has evolved from a collection of isolated techniques into a comprehensive, principle-based food production system. It strategically integrates conventional and indigenous farming knowledge with agroecological science, implementing practices such as cover cropping, agroforestry, green manure application, and minimal tillage. (Giller et al., 2021). Sustainable intensification has been defined as simultaneously raising productivity, increasing resource use efficiency, and reducing negative environmental impacts of agriculture (FAO, 2026b; Vanlauwe et al., 2023).



Second, revitalizing extension systems with consistent funding and testing innovative knowledge systems can enable farmers to adopt ISFM practices, including site-specific, customized recommendations, helping to catalyse the sustainable intensification of African agriculture. Strengthening advisory services can build farmers' capacity to innovate and develop ISFM approaches grounded in agroecological principles and recommended mineral fertilizer application rates. Implementing improved practices requires substantial training for technical staff to serve as facilitators, promote context-specific innovations, and mobilize scientific and local knowledge. These advisory services can be provided by the public sector, the private sector, NGOs, or farmers' organizations, through traditional extension, farmers' field schools, digital platforms, or other innovative approaches. Capacity development is often necessary regardless of the delivery model chosen. The Global Soil Doctors Programme in Africa is one example of a successful farmer-to-farmer training initiative that has demonstrated results in building farmer capacity for sustainable soil management (FAO and GSP, n.d.a).

Third, investment in research and development (R&D) at national and regional levels is another cross-cutting strategy for enhancing soil fertility and fertilizer use. To be effective, R&D efforts must be supported by comprehensive agricultural innovation systems that strengthen both research institutions and advisory services. Long-term planning is essential to set priorities in soil health, fertilizers, and agroecology research and innovation, as well as allocate human and financial resources accordingly. Policymakers should implement effective incentive structures to encourage private sector investment in agricultural innovation. This is key to creating a more robust and sustainable agricultural research ecosystem (Fuglie *et al.*, 2019), and integrating agroecological approaches into production systems. While agroecology offers long-term advantages, its widespread uptake by farmers, particularly those using traditional methods, is a gradual process that can be accelerated through targeted R&D and knowledge dissemination.

Fourth, financial institutions, including public development banks, are well positioned to mobilize investment for improving soil health and fertilizer use. They can do so by investing in large-scale soil testing facilities and ensuring that site-specific recommendations can be made across the widest possible range of territories, agronomic systems and farm households. Development finance institutions (DFIs) could offer provision of first-loss capital and the draw on African regional guarantee funds – such as the African Guarantee and Economic Cooperation Fund (FAGACE), the African Solidarity Fund (FSA), and the African Guarantee Fund (AGF). This strategic financial support helps to mitigate the high production risks associated with agriculture.

Fifth, robust monitoring, evaluation and accountability mechanisms are critical to ensuring the success of soil health and fertilizer initiatives. Continuous monitoring and learning from both past and current experiences enables verification that policy and programme objectives are being met. Effective oversight also serves as a safeguard, helping to prevent trade distortions and minimize opportunities for corruption. This function is best supported by long-term, multi-sector planning frameworks that incorporate national foresight processes – systematic assessments of future trends, risks and opportunities – to guide resource allocation and priority-setting over time.

Based on the issues presented in this report and the discussion of actions for achieving the four policy objectives, Table 2 presents an indicative list of potential policy instruments for enhancing soil health and increase producers' fertilizer use. Each instrument is assessed for technical efficiency, economic, environmental and social impact, and potential for scale. It is also referenced against the scientific evidence underpinning it. While the assessments reflect analytical insights, programme and policy impacts are ultimately shaped by the details and nuances of design and implementation, where political economy considerations are often decisive.

TABLE 2

Policy instrument options to enhance soil health and improve fertilizer use

OBJECTIVE	POLICY INSTRUMENT	TECHNICAL EFFICIENCY	ECONOMIC IMPACT	ENVIRONMENTAL IMPACT	SOCIAL IMPACT	SCALABILITY	REFERENCE
Improve soil health and soil fertility	R&D and support to marketing of bio-inputs	Potential high yield gains Not proven at scale in African contexts	Self-renewing Need confirmation of profitability	Positive / climate, soil, water	Potential pro-poor technology	No empirical information in Africa	Joshi and Gauraha, 2022 Dorin, 2022 Sandhu et al., 2023.
	Subsidies/ payments for environmental services of agroecological intensification/ increased use of integrated soil fertility management	Limited intensification in Africa Labour-intensive Long-term technical benefits	Need confirmation of economic impact	Positive / SOC	High labour requirement	Positive indications of effect on degraded soils. Scaling hinges on multiple factors	Wezel et al., 2020 Titttonell et al., 2022 Côte et al., 2022 Falconnier et al., 2023 HLPE, 2019
	Invest in localised soil laboratories for soil testing and promote side specific fertilizer recommendation	Tailored knowledge of nutrient application needs and soil structure improvement needs	Proven public good value	Less nutrients leaking	Improve producer productivity and income	Expected higher take up competitive advantage on degraded soils	Bekunda et al., 2005 Smaling and Braun, 1996
	Increase and optimize the use of bio-based fertilizers (incl. manure) in crops	Efficient in all African contexts	Labour constraint	Benefits / SOC but methane emissions	May not be socially beneficial; inter and intra-household issues	Limited availability of manure Need for external nutrients for feed	Bayu, Rethman and Hammes, 2005 Ndambi et al., 2019
Increase fertilizer availability	Invest in mineral fertilizer production plants and provide incentives for serving the local markets	Uncertain impact on food production depending on local demand for fertilizer/ foreign demand for fertilizer	High fixed costs	Carbon and NO _x emissions If locally used: nitrates but potential contribution to soil regeneration	Potential job creation with multiplier effects	Potential increased availability of fertilizer; Contribution to regional sovereignty	Snyder et al., 2009 Chien, Prochnow and Cantarella, 2009 Dangote Fertiliser, n.d.
	Invest in green hydrogen-based fertilizer domestic production	Uncertain impact on food production depending on fertilizer demand/ foreign demand for fertilizer	Very high fixed costs	Sustainable /oil carbon neutral NO _x emissions nitrates if local use	Potential job creation with multiplier effects	Potential increased availability of fertilizer; Contribution to regional sovereignty	Mostafaeipour and Sadeghi Sedeh, 2019 Hailes, 2023 Karimi Alavijeh, Yaghmaei and Mardanpour, 2020
	Invest in organic fertilizer production	Uncertain	Uncertain	Positive / SOC	Potential job creation with multiplier effects	Little evidence of a viable demand for bio-based fertilizers	Blanchard et al., 2013 Walling and Vaneeckhaute, 2020 Fadare et al., 2010 Assefa and Tadesse, 2019
	Invest in rural infrastructure (roads, storage, power grids, information systems, extension services)	Improved input access, improved output sales	Price stabilization Farm-gate food price increase Urban food price decrease	Depends on the infrastructure	Social benefits in remote areas	Scalability improving factor	Asher and Novosad, 2020 Le Cotty, Maître d'Hôtel and Ndiaye, 2017

Continued on next page

Increase access to and affordability of fertilizer	Subsidize mineral fertilizer	Limited impact on production, especially on degraded lands	Limited long term	Potential contribution to carbon and NO _x emissions and contribution to soil regeneration	Can be socially smart But risk of vested interests	Potentially scalable but in practice limited to some areas, or some groups of farmers	Jayne et al., 2013 Takeshima and Liverpool-Tasie, 2015
	Subsidize / develop the supply & marketing of bio-based fertilizer	Little evidence	Long-term benefits for land productivity & fertilizer profitability	Long-term benefits for soils	Can be pro-poor	Potentially scalable But little evidence	Epule et al., 2015 Chianu, Chianu and Mairura, 2012b
	Develop low risk credit for inputs	Potentially efficient especially in integrated value chains (e.g. cotton)	Positive if food dynamic markets Risky in remote areas		Depends on the interest rate & default risk	Could be scalable if strategies against default risk are improved	Adjognon, Liverpool-Tasie and Reardon, 2017 Mdoda et al., 2019
	Expand inventory credit	Improved access to fertilizer, income increase	Economically sustainable		Enhanced food security at the village scale	Limited to village scale	Delavallade and Godlonton, 2023 Le Cotty, Maitre d'Hôtel and Subervie, 2023
Increase producers' production incentives	Stabilize or support food prices	Very effective when properly administrated, in non-agricultural economies (many taxpayers, few farmers)	WTO incompatible if support is too strong and lasts too long	Depends on how affects production incentives	Price increase for the consumer Possible fiscal revenue if tariffs	National scale very demanding in terms of administrative skills	Barker and Hayami, 1976 Qian et al., 2020 Harvey, 2015
	Promote crop insurance	Imperfect indices, reluctance to the forward-looking requirement of insurance	Theoretical profitability		Social benefits for the insured	Scalable if subsidized low take-up in most countries	Ntukamazina et al., 2017 Fonta et al., 2018 Tadesse, Shiferaw and Erenstein, 2015

	Unambiguously positive
	Mixed literature
	Unambiguously negative
	Not applicable

SOURCES: Authors' own elaboration from: See References for Table 2 in References Section.

Conclusion

This report emphasizes the importance of improved soil fertility and soil health, increased fertilizer use, and greater fertilizer efficiency – grounded in “good agricultural practices” – as foundations for sustainable agrifood systems in sub-Saharan Africa. Historically slower productivity growth across the continent reflects broader agricultural challenges, in which inadequate fertilizer application and poor soil health are both causes and consequences of imperfect agrifood systems. While governments and regional groupings across the continent are collectively committed to improving food security through higher agricultural productivity, this report highlights the complex and multifaceted challenges that stand in the way. Overcoming these challenges requires more than just political commitment – it requires a coherent set of policies addressing the diverse factors affecting fertilizer use, soil health and fertility, both within and beyond the agricultural sector, backed by substantial investments that translate commitments into action.

Soil fertility functions as a conditional constraint: its importance for productivity growth is undeniable, but the degree to which it limits productivity varies considerably among farmers and contexts. Subsistence smallholders, who dominate SSA's agricultural landscape, cannot ease soil fertility bottlenecks without coordinated responses that simultaneously address risk, market incentives, land tenure, and infrastructure, alongside agronomic considerations. This does not reflect a failure of technology, but the presence of multiple, reinforcing constraints that shape farmers' decisions. The implication for policy design is direct: interventions that address only one constraint at a time – whether fertilizer access, extension, or soil restoration – will consistently underperform.

This report emphasizes the importance of improved soil fertility and soil health, increased fertilizer use, and greater fertilizer efficiency – grounded in “good agricultural practices” – as foundations for sustainable agrifood systems in sub-Saharan Africa.

Increased application of mineral fertilizer remains necessary across much of sub-Saharan Africa to raise yields in the near term, complemented by proven and financially viable agroecological and ISFM principles that build soil organic matter and nutrient availability over time. The need is clear in sub-humid and semi-arid parts of the continent and more fertilizers are also clearly needed in rainfed production systems. The case for greater fertilizer use is reinforced by evidence that fertilizers are on average more productive in SSA than elsewhere.²⁴ This is a consequence, in part, of the severity of soil degradation across the region, which makes investment in soil fertility and health more urgent and more impactful than in less degraded agrifood systems. Sustainable approaches tailored to specific soil characteristics and farmer circumstances are especially important in the most degraded areas, where blanket fertilizer application is least likely to generate durable returns.

This report argues that SSA countries do not face a binary choice between mineral fertilizer-led intensification and exclusively bio-based approaches. Most would likely benefit from a combination of complementary policy instruments supporting both. Central to any such framework is attention to the agronomic conditions of the soil itself – the key element of production. This implies a radical rethinking of fertilizer support programmes, moving away from blanket fertilizer subsidies toward more flexible, differentiated support. This approach needs to account for cropping systems dynamics, recognizing that intercropping, crop rotation, and crop–livestock integration are core soil fertility strategies for many smallholders, not a transitional step toward some other system. Recommendations and support programmes should be differentiated by farm size, agroecology, and market integration, and accompanied by investment in knowledge and behaviour change. Connecting production areas to dynamic consumption centres is equally important, with market integration not peripheral to soil health policy, but central to the profitability and sustainability of investments in production.

The policy instruments outlined in this report address potential trade-offs between short-term needs and long-term objectives. Short-term measures alone are insufficient to feed growing populations sustainably, while long-term

²⁴ Since they are a limiting factor in SSA, their marginal productivity is greater than if they were optimally used.



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strategies often face political obstacles. The ideal would be for countries to define their long-term vision for agrifood systems and design present policies in ways that move progressively towards it. The analytical foundations and policy tools to do this already exist – what is needed is the institutional commitment to apply them systematically.

The African Union and other regional economic communities continue to have a pivotal role to play in sustaining momentum for soil health improvement and sustainable fertilizer use. That role includes calling on member states to update and implement agrifood systems strategies, policies, regulations and budgets that invest in integrated soil health and nutrient management over the long term – encompassing both mineral and bio-based products, and treating soil as the foundation of food security, not just food production.

Annexes

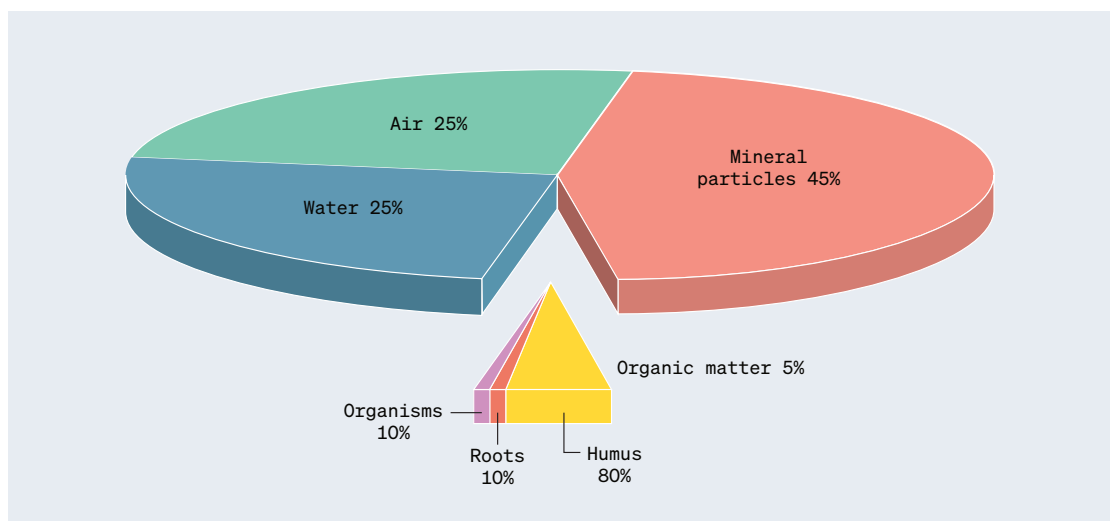
7.1

ANNEX 1: DEFINITIONS

Soil fertility is the ability of a soil to sustain plant growth by providing essential nutrients and favourable physical, chemical, and biological conditions as a habitat for plants (FAO and GSP, n.d.b). Nutrients reach the soil through several pathways – through inherited parent material, cycled through organic matter, or supplied through mineral or bio-based fertilizers – along with other growth factors such as air and water.

Figure 15 shows the typical composition of a healthy soil (Batjes, 2017). When these proportions are altered – through nutrient mining, inappropriate farming systems, inefficient fertilizer use or water stress – the health of the soil is compromised.

FIGURE 15
Composition of a healthy soil



SOURCE: Batjes, N.H. 2017. Benefits of soil carbon - an overview. Presentation at the GSOC2017 Digital Soil Organic Carbon Mapping Training, 6-23 June 2017, Wageningen. Wageningen, ISRIC & FAO/ GSP. www.isric.org/sites/default/files/GSOC2017mapping_BenefitsOfSoilCarbon_BatjesNH_2017.pdf

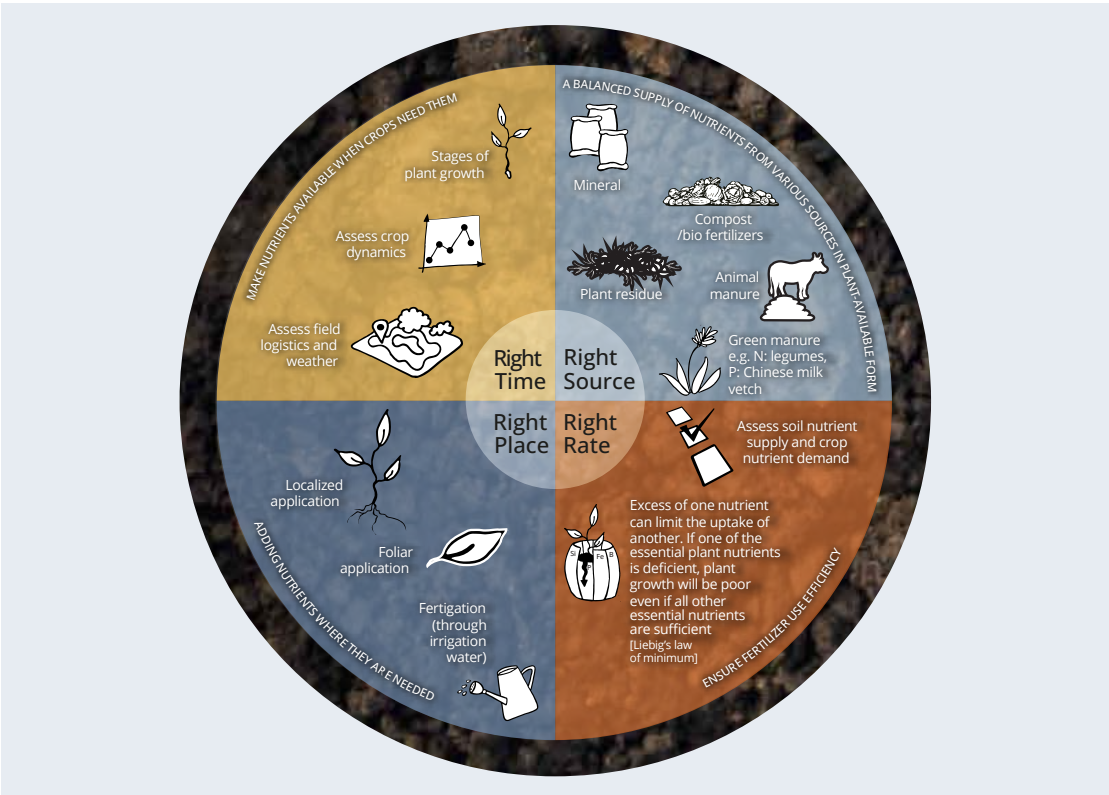
Soil health: The Intergovernmental Technical Panel on Soils (ITPS) defines soil health as “the ability of the soil to sustain the productivity, diversity, and environmental services of terrestrial ecosystems” (FAO and ITPS, 2020). While soil fertility refers specifically to the provision of nutrients and favourable growing conditions for plants, soil health is a broader concept encompassing the full range of physical, chemical, and biological properties that determine how well a soil functions as a living ecosystem (FAO and GSP, n.d.b).

Soil health is measured using a combination of indicators, including soil organic carbon (SOC), nutrient content, soil texture and structure, pH, microbial activity, and water holding capacity. These factors interact to create conditions favourable for plant growth and support a range of climate and environmental benefits. Nutrient content is thus only one aspect of soil health, and fertilizers are just one variable in the soil health equation.

Given that soil degradation involves both physical loss through erosion and nutrient depletion through uptake without replenishment, restoring and maintaining soil health requires more than fertilizer use alone. ISFM combines efficient use of mineral and bio-based fertilizers with appropriate soil and water management practices, including efficient farming systems, soil erosion control measures and improved crop variety selection (IFA, WFO and GACSA, 2016). Fertilizer use is guided by the 4R principles of nutrient stewardship – right source, right rate, right place and right time (FAO, 2023). Figure 16 illustrates these principles.

Cation-exchange capacity (CEC) is used as a measure of fertility, nutrient retention, and the capacity to protect groundwater from contamination by cations – positively charged ions such as Ca, Mg, Na, K and NH₄. CEC expresses the maximum quantity of cations that a soil can hold at a given pH value, available for exchange with the soil solution and so accessible to the roots. CEC is primarily associated with clay particles and humus; soils with low CEC –

FIGURE 16
The 4Rs stewardship principles of integrated soil fertility management



SOURCE: FAO. 2023. Technical guidelines on soils for nutrition – sustainable soil management for nutrition-sensitive agriculture. Rome. <https://doi.org/10.4060/cc5069en>

typically sandy or with little organic matter – have poor ability to retain nutrients. CEC is measured in centimoles of charge per kg of soil (cmol_c/kg) or milliequivalents per 100 g of soil (meq/100 g) (FAO, n.d.a).

Healthy soils offer better nutrient cycling and water retention, and support climate and environmental outcomes, mitigating erosion, controlling flooding, increasing biodiversity, cleaning water, and sequestering carbon (CASA, 2023). According to the Global Soil Partnership (FAO, 2022c), healthy soil is a dynamic living ecosystem composed of minerals and organic matter and containing air, water and living organisms. Soil health can be sustained or improved through well-managed nutrient supply, including the efficient use of fertilizers and appropriate practices and applications.²⁵

Soil degradation refers to a deterioration in soil health status that diminishes the capacity of an ecosystem to provide goods and services for its beneficiaries (FAO, n.d.b). It results from the alteration or depletion of one or more of the natural soil's physical, chemical, or biological properties – including its mineral content, organic matter, living organisms, air and water – with consequences for soil fertility and nutrient balance.

Principal causes include:

- soil erosion
- loss of natural vegetation, including forest
- soil compaction due to overgrazing and excessive use of heavy machinery
- desertification
- salinization, acidification, and alkalinization
- effects of toxic chemicals, pollutants, or excessive flooding and drought.

Tackling these causes requires a combination of interventions, and fertilizer use alone – whether mineral, bio-based or both – is insufficient to reverse soil degradation.

7.2 ANNEX 2: TYPES OF FERTILIZERS

This annex provides an overview of fertilizers used in SSA, including the dominant group of mineral fertilizer products (also referred to as “chemical” or “synthetic” fertilizers), as well as emerging alternatives such as green ammonia and bio-based fertilizers (see Table 3).

7.2.1 Mineral fertilizers

According to the FAO International Code of Conduct for the Sustainable Use and Management of Fertilizers, a fertilizer is defined as any substance that is used to provide nutrients to plants, typically applied to the soil, but also to foliage or through water in rice systems, fertigation, hydroponics or aquaculture operations (FAO, 2019). The International Fertilizer Association (IFA) specifies that mineral fertilizers are produced from naturally occurring nutrient deposits, or from the fixation of nitrogen from the atmosphere into plant-available forms (IFA, 2024). FAO further defines mineral fertilizer as a nutrient-rich fertilizer produced industrially by chemical processes, mineral extraction or by mechanical grinding (FAO and GSP, 2020; FAO, 2019).

²⁵ In the early 1990s, Europe began steps to control fertilizer use; nevertheless soil health remains a significant problem, with estimates of up to 70 percent of EU soils are in poor condition. Healthy soils are a central objective of the European Union's Soil Strategy for 2030, which forms part of the European Green Deal (Panagos et al., 2022).

TABLE 3
Fertilizer categories

TYPE	CATEGORY	EXAMPLES
Traditional	Mineral	A. Primary nutrients: - Single fertilizers: - Nitrogen fertilizers: mainly nitrate-based such as ammonium nitrate (AN), calcium ammonium nitrate (CAN) and urea and urea ammonium nitrate (UAN) most used in SSA. - Phosphorus fertilizers: monoammonium phosphate (MAP), single super phosphate (SSP), diammonium phosphate (DAP) and triple superphosphate (TSP). DAP and TSP are the most used in SSA. - Potassium fertilizers: potassium chloride (KCl), potassium sulphate (K_2SO_4), sulphate of potash (SOP), potassium nitrate (KNO_3) - Compound (mixed) fertilizers: - With two macronutrients: NP (e.g. diammonium phosphate) - With three macronutrients: NPK (e.g. 17-17-17) B. Secondary nutrients: e.g. calcium, magnesium and sulphur These are not usually applied as straight fertilizers but in combination with the primary nutrients N, P, and K. Both primary and secondary nutrients are the major nutrients useful for plant growth. C. Micronutrients: e.g. iron (Fe), manganese (Mn), boron (B), zinc (Zn), copper (Cu), molybdenum (Mo) and chlorine (Cl). Regulations recommend that the nutrient content always be shown on the fertilizer bag.
	Green ammonia	Produced via electrolysis, using renewable electricity. Hydrogen produced by electrolysis of water is combined with nitrogen from the air to create anhydrous ammonia – the same output as traditionally produced for mineral fertilizers.
Bio-based fertilizers	Organic	Animal manure, compost, food waste, crop residues, mulch, green manure, vermicompost, fish emulsion, bone meal, blood meal, biochar, bokashi.
	Biological products	Living organisms, including nitrogen-fixing bacteria (rhizobia), phosphate-solubilizing bacteria, and arbuscular mycorrhizal fungi.

SOURCE: Authors' own elaboration from CASA. 2023. *Exploring fertilizer alternatives and opportunities for improving use efficiency in Africa and South Asia*. Wellspring. London, CASA & Foreign, Commonwealth and Development Office (FCDO). <https://casaprogramme.com/wp-content/uploads/2024/06/Wellspring-October-2023-Exploring-Fertilizer-Alternatives-and-Opportunities-for-Improving-Use-Efficiency-in-Africa-and-South-Asia-CASA.pdf>

Mineral fertilizers can play a pivotal agronomic role by enhancing the growth of plants. They do this through providing nutrients necessary for plant growth –nitrogen (N), phosphorus (P) and potassium (K) among others. Mineral fertilizers generally contain high concentrations of a single, or two or the three nutrients, potentially combined with other nutrients. These three main macronutrients are especially important for leaf growth; development of roots, flowers, seeds and fruit; and strong stem growth, movement of water in plants, and promotion of flowering and fruiting (Dittmar *et al.*, 2009). Nitrogen is the most important, given its presence in proteins, DNA and chlorophyll. However, only a limited number of plant species – legumes in particular – can fix nitrogen from the atmosphere. Phosphate is important to produce DNA, ATP (the main energy carrier in cells), and certain lipids.

Two categories of mineral fertilizers are commonly in use (FAO and GSP, 2020):

- Fertilizers containing only one primary macronutrient, referred to as straight or single fertilizers.
- Fertilizers with two or three primary macronutrients, referred to as multi-nutrient fertilizers. These can be either compounds/complexes (all nutrients in the same granule) or bulk blends (physical mixing of different granules).

1. Straight or single fertilizer examples:

- Urea (46-0-0): containing 46 percent nitrogen
- Ammonia: containing 82 percent nitrogen
- Urea ammonium nitrate: containing 32 percent nitrogen
- Ground rock phosphate: containing phosphorus (40 percent P_2O_5)
- Triple superphosphate (0-46-0): containing phosphorus (46 percent P_2O_5)
- Potassium chloride (0-0-60): containing potash

2. Multinutrient/compound or mixed fertilizer examples:

- Diammonium phosphate (18-46-0): containing nitrogen (18 percent N) and phosphorus (46 percent P_2O_5)
- Potassium sulphate: containing potash (50 percent K_2O) and sulphur (18 percent S)
- Ammonium sulphate: containing nitrogen (21 percent N) and sulphur (24 percent S)
- Potassium nitrate: containing nitrogen (13 percent N) and potash (44 percent K_2O)
- Triple 17 (17-17-17): containing nitrogen (17 percent N), phosphorus (17 percent P_2O_5) and potash (17 percent K_2O)

7.2.2 Green ammonia

Green ammonia is produced via electrolysis, using renewable electricity. The hydrogen produced by electrolysis of water is then combined with nitrogen from the air to create anhydrous ammonia, which is the same output as traditionally produced via the Haber-Bosch process for mineral fertilizers. A key benefit is that despite being energy-intensive, the production of green ammonia can be carbon-neutral.

While most green ammonia production remains at a pilot stage, developments are accelerating.²⁶ The International Energy Agency (IEA) estimates that by 2030 nearly 8 megatonnes of near-zero-emissions ammonia production could be operational worldwide (IEA, 2021). Production is still in its infancy in sub-Saharan Africa, but the pipeline of planned projects is growing, with flagship initiatives in South Africa and Kenya (Hive Energy, 2021; Ammonia Energy Association, 2022).

Constraints for deployment include, first and foremost, the high capital costs of large-scale production and storage infrastructure. Electricity accounts for approximately 85 percent of the total cost of production (Royal Society, 2020), making access to affordable renewable energy essential for green ammonia development. A further constraint is that direct application of anhydrous ammonia as a fertilizer requires specialized knowledge, and equipment beyond the reach of most smallholder farmers in sub-Saharan Africa.

²⁶ This sub-section and the two that follow it draw on CASA (2023), which provides a clear and up-to-date description of alternative fertilizer products and market developments.

In practice, ammonia is most widely used as an intermediate product in the manufacture of other fertilizers, and its direct use is confined to very few countries – notably in North America – which account for 3–5 percent of global ammonia use.

Inefficient application of the resulting fertilizers can have negative environmental and health impacts, including water pollution and nitrous oxide emissions. Moreover, the production process also has a significant water requirement: electrolysis – the step producing the hydrogen used to make ammonia – requires large volumes of water.

In short, while green ammonia offers the prospect of substantially lower carbon emissions than conventional production, it faces considerable technical, economic and environmental hurdles before it can become a viable and sustainable fertilizer product (Santhappan *et al.*, 2023).

7.2.3 Organic products

Organic products are material derived from organic compounds from animals and plants that improve the physical, chemical and biological characteristics of soils, provide nutrients and increase microbial activity (Bullor *et al.*, 2024). Organic fertilizers include materials from livestock production operations used for fertilization purposes, such as manure, urine, slurry, straw and other bedding materials, (FAO, 2019). They also include compost (a mixture of decaying organic matter, as from leaves, domestic waste and manure, used to improve soil structure through the addition of carbon and provide nutrients), green manure (plants that are grown in order to provide soil cover and to improve the physical, chemical, and biological characteristics of soil), crop residues (residues from previous or other crops incorporated in the soil), mulch (plant material that is kept on the soil or added to the soil for soil and water conservation purposes), marine waste (organic fish amino acid and seaweed), biochar and bokashi. By and large, these products support long-term soil health by increasing soil organic carbon, improving soil structure and water holding capacity. They can also support microbial activity, which in turn can support nutrient cycling, enhance resistance against diseases and pests, and promote overall plant health and productivity (CASA, 2023).

The availability and uptake of organic products vary considerably across and within sub-Saharan Africa, because of variations in farming practices, the level of access to organic materials, local knowledge and traditions, policies, and farmer income levels. Organic products are often self-provided or obtained in local informal markets, resulting in a lack of official statistics. In most remote farming communities, organic “waste” is a misleading word as these resources compete with other uses and can be in short supply compared to demand.

Organic fertilizers play a critical role in ISFM and in improving all soil characteristics (physical, chemical and biological) and ultimately improve and sustain soil health. For that critical reason, they need to be promoted and strengthened by enhancing research and development to improve their quality and accelerate their production in Africa as reflected in the Africa Fertilizer and Soil Health Action Plan and the Nairobi Declaration (African Union, 2024b; African Union, 2024d).

For smallholder farmers in sub-Saharan Africa a blend of organic and mineral fertilizers – where available – may offer an optimal benefit/cost ratio in the future. A meta-analysis conducted in 2020 showed that combining

high-quality organic products with mineral fertilizers enhanced agronomic efficiency (the increase in yield per unit of nutrient applied) by 20 percent and reduced SOC losses by 18 percent across seven cropping seasons, compared with the sole use of mineral fertilizers (Gram *et al.*, 2020).

In regions of sub-Saharan Africa with particularly low use of mineral fertilizers, use of organic fertilizers is often the norm due to on-farm availability of animal manure and plant residues. For small-scale farmers, properly managing manure can make a significant contribution to soil health and crop yields improvement. While wealthier farmers can optimize nutrient uptake by using advanced storage methods, poorer farmers face challenges that lead to nutrient wastage (Rufino *et al.*, 2007). Evidence from the case studies reviewed by CASA suggests that the formal market for organic fertilizers is still nascent, with more anecdotes than evidence of profitable use at scale. However, the potential has spurred a range of efforts across Africa (see Box 3 in Annex 2).

Several constraints limit greater use of organic products. First, they are significantly less nutrient-dense than mineral fertilizers, and offer much less control and consistency of nutrient content across the manufacturing process. In addition, organic products may contain varying amounts of harmful agents, including weed seeds, plant pathogens, toxic chemicals, and antibiotics, which can enter the food chain. In particular, a range of bacterial, viral, and parasitic pathogens in organic fertilizers have been linked to foodborne outbreaks through crop contamination, posing risks to human and animal health (Adegoke, Awolusi and Stenström, 2016). Organic products are also commonly bulky, requiring larger volumes to be effective: this increases transportation, storage, and application costs per tonne, let alone the negative climate footprint of transport emissions. In addition, it can take several years to reap tangible benefits in terms of soil health, which can lead to losses for farmers in the short term.

In terms of sustainability, inefficient use carries some of the same risks as mineral fertilizers, including nutrient runoff (albeit to a lesser extent given lower nutrient concentrations), soil contamination (metal compounds, phthalates among other). The production and use of animal manure emits methane, which is a powerful greenhouse gas. Depending on the product and context, the impact of organic fertilizers can stimulate either nitrification (generally favourable as it converts nitrogen to a form that can be more readily assimilated by plants, potentially enhancing nitrogen use efficiency), or, inversely, denitrification (harmful under certain conditions, leading to increased production of nitrous oxide). This can be difficult to predict in advance, making the blanket prescription for “more” organic fertilizer products potentially counter-productive and requiring a quality control framework. In any case, the use of mineral fertilizers or organic fertilizers alone has some negative effects and ideally the combination of organic and mineral fertilizers as a climate-smart ISFM practice in sub-Saharan Africa has more beneficial effects. Results have shown that combined application of mineral and organic fertilizers leads to greater responses in productivity and agronomic efficiency as compared to sole applications of one of fertilizer types (Gram *et al.*, 2020).

Examples of emerging business models with the potential for scale at a local level include well-organized production of organic fertilizers.

These are produced in solid and liquid forms in various sub-Saharan countries: Benin, Burkina Faso, Nigeria, Senegal and Togo in West Africa; Eritrea, Ethiopia, Kenya, Malawi, Rwanda, Uganda, and the United Republic of Tanzania in East Africa; as well as in South Africa.

Mbeya organo-mineral fertilizer, produced by farmers in Malawi, is an example of a homemade organic fertilizer created from locally available raw materials combined with a small amount of inorganic fertilizer (FAO, 2025b).

Kenya is home to several initiatives, including waste-to-fertilizer and waste-to-biochar processes (CASA, 2023).

Eritrea has had success through a joint effort between the Ministry of Agriculture and the Ministry of Marine Resources to produce solid and liquid organic fertilizers, including bokashi fermented organic fertilizer, organic fish amino acid, and seaweed liquid fertilizer (Ministry of Agriculture, 2023). Eritrea has moved to establish regulations on quality standards for organic fertilizer exports as part of its collaboration between the two ministries to promote the production of these products (Ministry of Information, 2024).

7.2.4 Biological fertilizer products

Biofertilizers (FAO, 2019) are defined as products containing living or dormant microorganisms such as bacteria, fungi, actinomycetes and algae, alone or in combination, which on application help in fixing atmospheric nitrogen or solubilize/mobilize soil nutrients. They are living microorganisms that can improve soil health and crop yields.

Biological fertilizer products (also dubbed “biofertilizers”) are naturally derived substances and living organisms that can help to optimize nutrient uptake by soil and plants. They are used to supplement either mineral or organic fertilizers, rather than as a standalone input, and they help to transform nutrients present in the soil (or applied with organic or mineral fertilizer) into forms that are easily absorbed by plants.

The distinction is usually made between biofertilizers and biostimulants (including seaweed extracts and humic acids, which work to improve fertilizer efficiency). Biofertilizers are typically delivered via seed treatment (coating seeds before planting), soil treatment (mixing into the soil), or root dipping (for transplanted crops). The method that is chosen depends on the crop, biofertilizer type, and field conditions.

A scant but encouraging set of case study evidence, including from sub-Saharan Africa, shows positive outcomes in terms of improved yields of up to 30 percent versus untreated crops (Schutz *et al.*, 2018). One of the constraints for deployment lies in the timing, frequency, and method of application, which are critical issues that should be tailored as far as possible to specific crop and soil conditions. In addition, specific storage conditions – in some cases refrigeration – are needed to preserve the efficacy of living products, which can be

difficult to ensure during “last-mile” distribution in parts of sub-Saharan Africa and represent additional logistical costs depending on origin, packaging, and shelf life among others, leading to a high cost for the farmers to incur. Further, development of biofertilizers in the region would require development of policies, regulations, and rules for quality control (Raimi, Roopnarain and Adeleke, 2021). The largest customer segment to date is large commercial farms in Europe, the United States, and other developed economies, with smallholder farmers in sub-Saharan Africa still only a nascent customer group (Raimi, Roopnarain and Adeleke, 2021), with early-stage development of biofertilizers in countries such as Côte d’Ivoire for cotton production.

Currently, research on use of biofertilizers in several crops is ongoing in Africa (AATF and IITA, 2013). These include for sorghum in Ethiopia, Kenya and Zimbabwe; for cowpea in Cameroon; for groundnut and bambara groundnut in Madagascar; for rice in Rwanda; and for unspecified work in Burkina Faso, Côte d’Ivoire, the Democratic Republic of the Congo, Kenya, Rwanda and Senegal. The UNESCO Microbiological Resources Centre (MIRCEN) project at the University of Nairobi, Kenya has, since 1981, developed a *Rhizobium* inoculant known as BIOFIX, which is the main inoculant currently available on the local market and therefore used by farmers.

Under the right conditions, biological fertilizer products have the potential to reduce the need for mineral fertilizers (by increasing the efficiency of the fertilizer or enriching the soil with nutrients available to plants), reducing the associated emissions and soil and water pollution (Saffeullah *et al.*, 2021).

7.2.5 Organo-mineral fertilizers

Organo-mineral fertilizers combine dried organic and mineral fertilizers to provide balanced nutrients along with soil health improvements in a long-lasting, easy-to transport and store form (IFA, 2024).



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Agriculture is a cornerstone of sub-Saharan Africa's economy, yet it falls short of meeting regional food security and nutrition needs. Low productivity is widespread, particularly where soil quality and health have been degraded. Fertilizers can play a key role in breaking this cycle, but only where conditions support their effective use. This report by the FAO Investment Centre, in partnership with Agrinatura and the European Union, examines the structural, market and agronomic constraints to improving soil health, and assesses the tools and options available to address them. It finds that raising yields will require increased mineral fertilizer use, complemented by proven agro-ecological and soil health management approaches. The evidence points away from uniform solutions and traditional approaches that have favoured fertilizer subsidies to limited effect, and shows that policies and programmes must be tailored to the agronomic realities and socioeconomic circumstances of different smallholder contexts. This report is part of the Directions in Investment series under the FAO Investment Centre's Innovation and Knowledge for Investment (IK4I) programme.

ISBN 978-92-5-140746-2



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CD9916EN/1/06.26